PRIVOKE: A Client-Side Privacy Firewall for LLM Prompts

Nicholas Bisset*
Monash University Clayton, Australia
nbis0006@student.monash.edu

Manpreet Sahadevan, Nixon Ngai, and Sanoop Mallissery
Monash University Malaysia, Malaysia
{msah0018, nnga0004}@student.monash.edu, sanoop.mallissery@monash.edu

***Abstract*—Useful prompts often contain information that users may not intend to disclose to an external language-model provider. PRIVOKE inspects supported requests locally and applies explicit allow, warn or block actions before transmission. A separate training path checks version and retention constraints before distributing accepted parameters, while independently randomized telemetry provides no training data. Across 1,560 authored cases with provisional targets, updated packages improved contextual policy agreement. Targeted detector removal nevertheless exposed a tradeoff between unnecessary restrictions and missed protection, and a frozen 498-row semantic annotation-presence check yielded specificity of only 13.36 to 21.76%. Thirty-three installed receiver probes verified the declared request behavior. Every matched full-model candidate improved aggregate agreement but failed retention. Installed pretrained-model measurements isolated resident detector-activation cost, although a quality-tier probe blocked a benign prompt. A detector-layer precision revision moved regex specificity on a previously opened endpoint, scored once for the detector layers, from 98.5 to 99.6% at a recall cost of 1.3 points; each shift rests on three rows and is not statistically significant (exact McNemar $p = 0.25$), the rules-and-NER union showed no detectable change (which does not establish equivalence), and no net NER benefit was established. A full-product rerun with these revised detectors reduced the 139 shared unnecessary restrictions to 108 without new binary misses, while the removal tradeoff persisted. Registered tests on fresh annotated sentences then qualified the revision at the detector layers: the first revision raised union specificity by 3.3 points but lost 1.37 points of union recall against the original detectors (interval -1.95 to -0.79), failing its non-inferiority test. A rework of the rules that cost recall was shipped by owner decision. On a second fresh draw it still lost 1.05 points of union recall (-1.70 to -0.41) while keeping BLOCK-level recall level, so it did not meet the registered promotion rule; the loss concentrated in one source and in the NER layer. Controlled telemetry tests verified delivery and accounting; separate audits identified a small excess over nominal privacy loss and substantial synthetic estimation error. Together, these findings demonstrate an inspectable protection and revision mechanism while leaving independent contextual accuracy, adaptation without lost protection, real-user monitoring usefulness and practical user benefit unresolved.**

***Index Terms*—Local differential privacy, local inference, model regression, prompt privacy, randomized telemetry, request enforcement.**

I. INTRODUCTION

Web-hosted large language models (LLMs) support writing, coding and analysis by processing users' instructions and contextual material [1]–[3]. That context can extend well beyond the immediate question. A person seeking help with a report, for example, might include a colleague's details, health information or confidential organizational material. Information that helps the model complete the task can therefore also disclose something the user intended to keep private. Reported workplace incidents and company warnings illustrate this risk during legitimate work [4], [5]. Controls on subsequent storage or use cannot prevent the initial transfer, whereas inspection on the user's device offers an opportunity to intervene before local control is lost.

Local execution of the application model is one alternative to sending a prompt to a hosted service. Research releases such as LLaMA demonstrate that foundation models can be made available for separate deployment [6], but moving the whole task locally changes the chosen application and its computational requirements. PriVoke instead studies intervention within an existing hosted workflow: a compact local detector decides whether a supported request may proceed to the external model. This choice makes detector cost, endpoint coverage and the consequences of interruption relevant alongside classification accuracy. It does not assume that hosted models are universally more capable than local alternatives.

The point of intervention also determines which privacy risks the system can address. Priyanshu et al. investigate the reproduction of sensitive prompt information in chatbot outputs and the effect of instructions intended to sanitize those outputs [7]. Such work concerns information already supplied to the model. Pre-transmission inspection can prevent a particular submission, but cannot recover a previous disclosure or protect information held by a provider through an unrelated channel. The motivating workplace examples therefore establish a reason to examine outgoing disclosure, without making PriVoke a remedy for every platform-side privacy failure.

Deciding when to intervene involves more than recognizing names or identifiers. Prompts about the same subject can differ in whose information they contain, their intended recipients and the appropriateness of disclosure. Nissenbaum's account of contextual integrity likewise relates privacy to the norms governing information flows within a particular context [8]. Detector labels can represent some of these distinctions without establishing that a disclosure is authorized. The resulting decision must also affect the outgoing request: alerting a user

*Corresponding author: Nicholas Bisset.
GitHub Repository Link.

to sensitive content does not necessarily prevent its transmission. An evaluation must therefore distinguish contextual predictions, policy decisions and what actually leaves the device.

This distinction shaped the variables used in the research. Sensitivity describes the information's potential for harm; intended audience describes the claimed disclosure context; and information categories identify overlapping types of content. The requested detector layers and the order in which they execute determine which findings reach the policy. ALLOW, WARN and BLOCK then determine the request disposition. Treating these variables separately permits a useful diagnosis: a system may recognize sensitive content but choose an inappropriate action, or choose the intended action while describing the content incorrectly. Neither outcome can be inferred from an entity-presence score alone. Model capacity, input windows, pretraining and decision thresholds may also affect the balance between coverage, restriction and local cost. The compact-model precedent [9] motivates examining such configurations, but the native-package comparison changes several together; only the matched comparisons can narrow the attribution.

We address these connected decisions through PRIVOKE, an implemented client-side privacy firewall for supported outgoing prompts. It combines rules, named entity recognition (NER) and semantic classification to characterize sensitivity, intended audience and information categories, then selects ALLOW, WARN or BLOCK. ALLOW and WARN release the original request; BLOCK cancels it. WARN attempts to display a notice but does not wait for acknowledgement, so the warning itself demonstrates neither consent nor prevented disclosure. This explicit request contract makes it possible to assess classification and enforcement as related, but distinct, outcomes.

Maintaining the detector raises a further difficulty: a revision can improve average accuracy while becoming incorrect on cases its predecessor handled correctly [10]. PriVoke therefore separates learning a candidate from accepting it. A server-side runtime computes a candidate update from supplied labelled examples; the candidate is gated on declared retention criteria before the update service applies it only to its exact parent model. Clients can verify accepted artifacts and refresh local parameters without an application release. Reporting follows a separate path, with coarse event fields randomized on the workstation. Neither intercepted prompts nor telemetry supplies training labels, keeping maintenance and reporting independent of user traffic as a source of supervision.

These separations also constrain what adaptation can mean. The model's base version, trainable parameters, exposure to labelled examples and retention cases are experimental variables, rather than evidence that a revision will help. A changed input distribution may motivate supplying new examples, but a lower training loss cannot demonstrate preserved protection. Operational reporting introduces a different set of variables: which events are admitted, how their fields are randomized, the privacy budget consumed and the error in the resulting estimates. Reporting can be implemented correctly while failing to represent the events or users of interest. We therefore evaluate maintenance and reporting as distinct mechanisms, with no automatic learning loop inferred from their coexistence.

Casper, Hide and Seek and ProSan establish substantial precedent for local protection and learned sanitization [11]–[14]. Our contribution lies in the integration and empirical analysis of request enforcement, guarded updates and separate reporting; local detection, regression analysis and randomized response are established techniques.

Contributions. Our contributions are as follows:

1) A client-side semantic privacy firewall that combines local rule-, NER-, and semantic-based inspection with explicit ALLOW, WARN, and BLOCK enforcement before supported prompts leave the workstation.
2) A guarded model-update lifecycle that binds candidate updates to their parent versions and applies retention checks before publication, showing that aggregate accuracy gains do not necessarily preserve previously protected behavior.
3) A privacy-bounded telemetry mechanism that releases only randomized coarse event information under a local privacy-budget ledger, while remaining separate from user-prompt inspection and model training.
4) An end-to-end empirical evaluation of detection, request enforcement, model-update retention, runtime overhead, and telemetry privacy, revealing trade-offs between improved agreement, unnecessary restriction, lost protection, and monitoring utility.

We organize the evaluation around three questions:

1) **RQ1**: Does PriVoke detect and enforce privacy-sensitive prompt policies effectively?
2) **RQ2**: Can PriVoke update its detector without sacrificing previously protected behavior?
3) **RQ3**: What systems and privacy overhead does PriVoke introduce?

RQ1 examines the relationship between contextual prediction and enforcement, including situations in which better classification or fewer restrictions do not mean better protection. RQ2 tests whether aggregate improvements satisfy a casewise retention requirement. RQ3 measures systems costs while distinguishing report-content privacy from estimation error and contribution selection. Together, the questions establish what successful execution, training and accounting can demonstrate, and where the evidence remains limited by provisional contextual labels.

II. RELATED WORK AND CONTRIBUTION

Local intervention and contextual decisions. Casper places prompt protection on the user's device through a browser extension combining rules, entity recognition and local topic analysis [11]. In the reviewed prototype, a keyboard shortcut initiates inspection; identifiers are replaced with placeholders, users can request their restoration in responses, and topic warnings require acknowledgement. Custom rules, topics and compatible replacement models make the protection configurable.

PriVoke adopts the motivation for layered local inspection but implements a different request contract: supported hooks apply decisions automatically, and WARN releases unchanged content without waiting for a user response.

Hide and Seek (HaS) seeks to preserve useful interaction by hiding entities before cloud inference and recovering them in the response [13]. Its generative variant trains hiding and recovery models, whereas its label-based variant uses an entity recognizer for hiding. ProSan selects word replacements through task-importance and privacy proxies, then trains a sanitizer for local execution [14]. PromptShield approaches enterprise protection through role/policy classification [15]. Together, these systems show why detecting information, controlling its disclosure and completing the user's task warrant separate evaluation. PriVoke's labels describe its policy, but cannot by themselves validate contextual authorization.

Retaining behavior during model revision. Model-update research distinguishes improved average performance from preservation of individual predictions. Yan et al. examine *negative flips*, in which an updated image classifier becomes incorrect on a previously correct example, and propose positive-congruent training to reduce these regressions [10]. This motivates measuring both gains and losses, although image-classification findings do not establish privacy-policy correctness. PriVoke applies the distinction to publication: it compares labels and actions against the declared base and can reject an improved candidate through a configured retention guard. Linking this decision to a versioned artifact and verified client refresh separates successful optimization from acceptance for use. The evaluated connection is the contribution; the guard remains limited to its declared cases and criteria.

This framing does not require earlier systems to lack configurable or learned components. Casper permits custom rules and topics, and HaS and ProSan explicitly study learned transformations. The narrower question is how to connect a proposed classifier change to an identifiable parent, a declared behavioral acceptance rule and the artifact that would be served. That question remains meaningful even when the local detector, learning algorithm and publication machinery each use established techniques. It also makes a rejection informative: an update can reveal a useful learning direction while remaining ineligible under the chosen harm criterion.

Privacy of operational reporting. Local randomization supports aggregate collection without directly releasing each underlying value. RAPPOR applies randomized response to client-side statistics and addresses the additional privacy cost of repeated reporting [16]. PriVoke instead combines categorical event reports with a local budget ledger, without adopting RAPPOR's memoization. Its evaluation must consequently establish both whether the implemented sampler and accounting meet the stated event-content bounds and whether the estimates are useful. We examine finite arithmetic, estimation error and contribution selection separately because a content bound protects neither participation nor network metadata. Nor do the reported fields provide corrected contextual labels or establish a need to retrain the detector.

Contribution and comparison boundaries. These strands address different outcomes: sanitization and recovery concern disclosure and continued use, negative-flip analysis concerns changed predictions, and randomization constrains released observations. RQ1 connects layered decisions to receiver-observed transmission, without claiming Casper's entity/topic, HaS's recovery-resistance or ProSan's task-utility results. RQ2 distinguishes aggregate improvement from eligibility for version-bound replacement, while RQ3 examines cost, finite content privacy, estimation and admission selection. Making these decisions jointly inspectable reveals where they diverge. Their different targets and denominators, however, provide no basis for a numerical superiority claim.

III. System Design and Privacy Boundary

A. Scope and Trust

PriVoke intervenes when personal, third-party or organizational information is about to leave the workstation in a supported prompt request. It trusts the enabled extension, local runtime and supervisor, workstation, and authenticated publisher. Protection therefore depends on that boundary: compromised or deliberately replaced page hooks can bypass inspection, and blocking a later request cannot recover information already disclosed.

The interception path recognizes prompt POST formats sent through page `fetch` and asynchronous `XMLHttpRequest` (XHR). Workers, attachments, other transports and changed endpoints need separate validation. Users may also disable protection, deliberately bypassing inspection. An optional hosted-inference configuration moves text beyond the workstation and changes the disclosure boundary; it is consequently excluded from this paper's local-inference measurements.

Fig. 1 distinguishes prompt inspection from training on supplied examples and randomized reporting. A permitting action releases the prompt to the hosted LLM, but maintenance need not collect that prompt and the collector supplies no training labels. This separation defines the information paths without establishing that the resulting decisions are appropriate.

B. Local Inspection and Request Enforcement

The page hook extracts supported prompt text and passes it through the extension background and loopback gRPC-Web remote-procedure-call bridge to the workstation's Python runtime. Native messaging starts the supervisor that manages this path. The returned action determines whether the original request is released or cancelled; the separate server-side runtime instance that computes training updates never serves as an inference fallback.

The runtime normalizes text while retaining offsets linking findings to the original input. It produces two views with the same normalization procedure, run once per view: case-preserved text for the regular-expression and NER layers, and lowercased text for the semantic models, which were trained on lowercased normalized input. Unicode normalization, whitespace and obfuscation handling are shared, so both views map findings to the same original intervals and

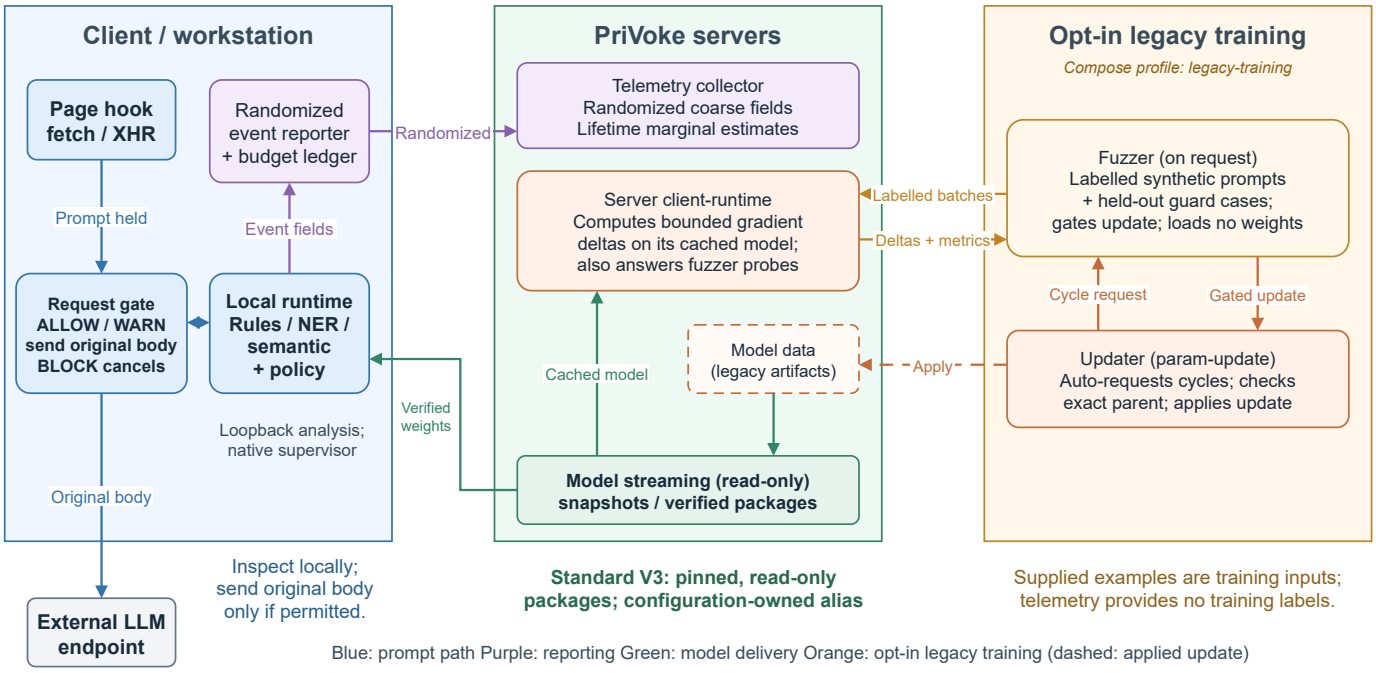


Fig. 1. PriVoke's client, server and opt-in legacy training paths. Local inspection returns an action: ALLOW and WARN send the original body; BLOCK cancels the request. Randomized coarse event fields pass through a separate budgeted reporting channel. Standard V3 uses pinned read-only packages. In the legacy route, the updater requests a cycle; the fuzzer generates labelled examples, the server runtime computes bounded updates on its cached model, and the fuzzer forwards only candidates passing its held-out guards. The updater applies an update only to its exact parent; this dashed path does not update standard V3.

differ only in letter case. The rule and NER layers also apply the precision controls of Section III-C. The runtime evaluates regular-expression rules before enabled NER and semantic layers, ending the cascade immediately if a rule returns BLOCK. Otherwise, it combines findings using the strongest action. As a result, a semantic classifier cannot reverse an earlier rule-based restriction, and an early BLOCK prevents the remaining requested detectors from running.

Failure handling also depends on which layers actually execute. A failed requested layer turns an otherwise ALLOW result into BLOCK, while valid non-ALLOW findings remain visible. The extension blocks supported requests if analysis itself is unavailable. Loss of the parameter stream can instead leave rules and NER running in a reduced configuration, which may release a request without semantic inspection. The action and execution record must therefore be interpreted together.

Semantic classification describes sensitivity, audience and information categories. Sensitivity ranges from no identified risk (S0), through low-risk personal context (S1) and potentially harmful personal information (S2), to highly sensitive information, identifiers or secrets (S3). Audience, termed *visibility* in the implementation, distinguishes public (P0), semi-public (P1), authentication-restricted (P2), group-private (P3), personal-private (P4) and unspecified (PU) contexts. Categories may overlap and cover health, politics, religion, criminal, financial, sexual, child, location, identity and third-party information.

The policy ordinarily maps S3 to BLOCK and S2 to WARN. S0/S1 allow unless identity and location occur together, or either occurs with visibility P2–P4; these exceptions produce WARN. S0 without categories always allows. Numeric confidence below 0.5 overrides this mapping: S3 becomes WARN, S2 remains WARN only when identity or location is present, and other cases become ALLOW. Confidence summarizes sensitivity and category heads rather than a calibrated probability of joint correctness or authorized disclosure. Neither the labels nor their deterministic mapping has been independently validated for contextual authorization. Appendix A states the decision order explicitly.

Consider a hypothetical S2 prediction with HEALTH, confidence 0.7 and no earlier restriction. The policy returns WARN, allowing the supported request to send its original body without acknowledgement. If a rule had already returned BLOCK, the request would instead be cancelled before semantic inspection. This example illustrates the mapping; it adds no observed prediction, independently labelled case or test.

These models classify prompts rather than generate conversational responses. Historical V1 packages use randomly initialized frozen encoders with fitted heads; V3 packages retain pretrained Bidirectional Encoder Representations from Transformers (BERT) encoders and poolers[1] [9]. Both families

[1]BERT L-4/H-128/A-2, revision c29bee83fc7f003ac8c5e6e135529da4ecddb7c3; BERT L-2/H-256/A-4, revision 4e937a8675e5afd9a4836735c186ec01695bc3ea; BERT L-12/H-256/A-4, revision 2ab9a0f41435a4d23d4cbc11cc3e7c922545f13d.

offer efficient, balanced and quality packages, whose names imply no guaranteed performance ordering. The installed default now selects V3 balanced, alongside explicit V3 tiers and labelled older V1 choices. Its CPU loader verifies binary packages from the model service without falling back to a random or V1 model. Differences in tokenizers, capacity, thresholds and training history mean that comparisons assess the complete package change, not pretraining or architecture alone.

C. Detector Precision Controls

Rules and NER run first, and a rule BLOCK ends the cascade, so their false detections cannot be repaired by a later semantic layer. Three controls therefore limit how often these layers fire without personal context. They change detector findings only; the policy mapping, the semantic models and the update path are unchanged.

Rule gates. A narrative heuristic that matched on length alone was removed. Kinship, family and child cues now require a possessive determiner, optionally with modifier words, or an age expression combined with a person noun. Financial keywords cover personal financial-state nouns after a possessive, and location keywords require a connector followed by a name rather than a closed-class word. Structured identity values remain on one short line. Rules derived from Presidio patterns regained validators: Luhn checksums [17] for card numbers, mod-97 for IBANs [18], mod-11 for NHS numbers, issuable ranges for US social security numbers and whole-word context terms for weakly shaped licence and date patterns. Placeholder passwords, chat mentions and standards identifiers are excluded. Appendix F lists the gates.

Labelled identifiers and checksum downgrades. Validators and the whole-word gate on the weak licence pattern also removed real identifiers: values that fail a check digit, follow a format no pattern recognizes, or are written without a context term. A later rework therefore added two general rules and a downgrade path. A value of identifier shape (at least one digit, five to twenty-four characters, not a date) introduced by an explicit label is treated as an identifier whatever its format, because the label, not a checksum or country pattern, is the evidence. A label is a qualifier plus an identifier noun ("passport number", `customer_id`), a self-labelling document ("Passport:") or an identifier acronym such as SSN. The qualifier class sets sensitivity: government, health-insurance and payment qualifiers yield S3, while customer, account, member, employee, user, policy, claim, registration, plate and store-card qualifiers yield S2. A failed checksum no longer drops the hit when context supports it. A failed Luhn check beside a card word, a failed mod-97 check beside a banking word and a failed social-security range beside a social-security word stay S3 under an "unverified" rule name; a long number-like token that fails the IBAN check (eight or more characters, five or more digits and a letter) is kept as S2; short or mostly alphabetic tokens and unlabelled digit runs are still dropped. Kinship, family and child cues were reviewed and left unchanged: no development row showed a missed first-person parenthood statement, and restoring bare kin or child nouns would bring back hits attributed to the wrong category.

Case-preserved input. All rules are compiled case-insensitively, so letter case does not determine which rules match. The small English spaCy NER model [19], however, uses capitalization as evidence for proper nouns, and lowercasing the input suppresses that evidence. Passing case-preserved text to NER therefore increases the entities it returns, but it also admits capitalized non-names such as headings, acronyms and ordinary words, which motivates the filter below.

NER post-filter. A first stage removes entity spans that are unlikely to be personal: masking placeholders, identifier-shaped tokens, spans inside structural headings, short acronyms without a locating or affiliation preposition, and single-token person candidates that an offline lexicon and tagger read as ordinary words. A second stage assigns sensitivity from the combination of surviving categories, following a quasi-identifier principle: a lone place or organization does not identify the speaker and is dropped unless a relation cue such as "I live in" ties it to the speaker; a person linked with a corroborated place or organization, or a corroborated name (multi-token, cue-introduced or recurring), yields S2; and an uncorroborated single name yields S1. The filter records the deciding rule for each emitted entity and uses only resources installed with the model.

The rule repairs and the labelled-identifier rules ship unconditionally. The case-preserved input and the post-filter are on by default and can be switched off individually (`PRIVOKE_CASE_PRESERVING_DETECTORS` and `PRIVOKE_NER_POSTFILTER`, set to 0), which preserves the pre-change behavior for ablation and rollback. A later action-level check (Section V-C) found that removing the child cue had downgraded identified adult student records from BLOCK to WARN. A student-identifier rule (S3 identity; a value of four or more characters containing a digit tied to the word student) was therefore added after the first evaluations were scored.

Three configurations are named in what follows. O is the pre-change detector set. N is the first revision: rule gates, validators, case-preserved input, post-filter and the student-identifier rule. K is the shipped configuration: N plus the labelled-identifier rules and checksum downgrades. The tables of Sections V-C and V-D measure the first revision before the student-identifier rule and the rework existed; Section V-E measures N and K on fresh data. Default-off input canonicalizations (stripping zero-width characters, rejoining split digits, spelled e-mail separators and a caseless gate on NER) and alternative NER models were also evaluated, but they are not part of K (Section V-F).

D. Guarded Detector Updates

Before accepting a candidate, the update path checks both its model identity and its behavior. In the legacy route, the updater periodically requests a cycle from a fuzzer, which prepares bounded labelled batches and held-out guard cases, excluding protected rows, texts and groups from training.

The fuzzer never loads model weights. Instead, the server-side runtime computes bounded parameter changes against its cached copy of the specified base, obtained from the model-streaming service, and returns them with base and candidate metrics bound to that base. The fuzzer forwards only candidates satisfying its configured quality and retention guards. The updater then checks parent identity, permitted tensors and shapes, finite values and magnitude limits before applying the update. Artifact checks establish that a candidate is well formed; retention checks assess its behavior on declared cases. Neither successful fitting nor parameter transfer establishes acceptance.

Publication uses atomic compare-and-swap: the current base must still match the candidate's declared parent. This prevents a decision about one model from replacing an intervening revision. Durable replay receipts connect identical retries to their earlier artifact outcomes, avoiding duplicate application. Head and underlying-model stages are separate publications, however, so rejection of a later stage does not undo an accepted head update. Clients verify delivered artifacts before refreshing cached weights. Standard V3 instead uses a pinned read-only package store and a configuration-owned alias, beyond the separately enabled legacy training/publication route. Unsupported V3 legacy-gradient calls are rejected; installed V3 inference thus provides no evidence of deployed V3 learning or guarded publication.

Training is centralized and uses supplied examples, without user gradients or federated learning. The implemented service path establishes neither cloud deployment nor the value of a particular revision.

E. Separate Randomized Telemetry

Telemetry reports five coarse event fields: action, an action/sensitivity risk bucket, one category chosen by fixed precedence, a bounded release label and a four-hour time-of-day bin without a date. Their domains have sizes $K_i = 3, 4, 11, 2, 6$, including a no-category value. Each field is randomized locally; raw prompts, identifiers, exact confidence and per-layer findings are omitted. This reduces disclosure without making the remaining fields harmless. The collector estimates lifetime marginals for usage and development inspection, but cannot estimate latency or error rates: it has no analysis/user denominator, date/window query or joint-event record, and distinguishes only v0.3.0 from OTHER. A separate Health call checks store writability. Neither channel produces corrected training labels.

Local differential privacy limits how much an output probability can vary between possible event contents [20]. This comparison fixes the reporting opportunity and public configuration, leaving the presence of a report visible. The event budget ε is divided equally across fields as $\varepsilon_i = \varepsilon/5$. For a domain of size K_i, the true value has reporting probability p_i and each alternative has probability q_i:

$$p_i = \frac{e^{\varepsilon_i}}{e^{\varepsilon_i} + K_i - 1}, \quad q_i = \frac{1}{e^{\varepsilon_i} + K_i - 1}. \tag{1}$$

Under independent ideal sampling, composition multiplies the field likelihood ratios to bound the event ratio by e^{ε} [20]. The implementation compares a fresh uniform 53-bit integer draw, $J/2^{53}$, with a binary64 threshold $\widetilde{p}_i$, choosing alternatives by rejection sampling. The resulting finite probabilities and event loss are

$$\begin{aligned} P_i &= \frac{\lceil 2^{53}\widetilde{p}_i \rceil}{2^{53}}, \quad Q_i = \frac{1 - P_i}{K_i - 1}, \\ \varepsilon_{\text{finite}} &= \sum_i \log \max\{P_i/Q_i, Q_i/P_i\}. \end{aligned} \tag{2}$$

The calculation depends on host arithmetic and independent uniform random bits, so the ideal formula alone cannot certify the sampler. A persistent ledger reserves budget before report construction, defaulting to one per event and eight per Coordinated Universal Time (UTC) day. Failed construction or delivery does not refund that reservation; accounting therefore requires retained state and a trusted clock. Report presence, contribution selection, delivery and network metadata fall outside the content guarantee. The collector estimates marginals by inverting nominal response probabilities and clipping counts to their feasible range, leaving their usefulness after randomization and clipping to empirical assessment.

IV. Evaluation Method

A. Questions and Evidence Units

Table I links the three research questions to evidence about prediction and enforcement, retained behavior, and systems and reporting costs. Because several analyses reuse cases, agreement between their findings does not constitute independent confirmation.

Product calls request rules, NER and semantics [2,3,4], recording any legitimate skips after a rule-based BLOCK. Sole-semantic calls explicitly request [4] and assert that it alone executes successfully. Detector-layer calls for the precision study request [2,3] and assert that both executed successfully. This distinction prevents product outcomes from being attributed to semantic classification alone, and detector-layer outcomes from being read as either.

The unit of analysis changes with the question. A contextual case is one authored situation with provisional targets, regardless of how many packages process it. A final annotation row is one frozen source example. A receiver probe is a specified request-handling scenario, while a browser timing observation is a repeated execution nested within a profile. A training seed identifies a fit from the common base, and a telemetry repetition samples reports for a fixed contributor histogram. We retain those units when interpreting counts and uncertainty; execution volume does not turn repeated observations into additional independently sourced cases.

Contextual targets and measures. The authored workload contains 1,560 cases, comprising 810 sensitive and 750 clean examples, with 260 per audience class, 35 scenario families and 24 audience-expression groups. Targets for sensitivity, audience, categories and allowed actions were prepared provisionally with assistant support, without independent human

TABLE I
EVIDENCE LINKING THE RESEARCH QUESTIONS TO THEIR MEASURED OUTCOMES. CONTEXTUAL TARGETS ARE PROVISIONAL AND DO NOT CONSTITUTE INDEPENDENTLY ADJUDICATED PRIVACY TRUTH.

RQ	Evidence	What it establishes
1	1,560 contextual cases; targeted ablation; 498 final presence rows	Agreement and detector contribution on declared targets.
1	Detector-layer precision on design, check and held-out splits	Regex/NER specificity and recall before and after the precision revision, with paired tests, without product actions.
1	Full-product rerun of the 1,560-case comparison and ablation	Whether restriction and removal findings persist with the revised detectors.
1	Registered fresh held-out detector-layer draws; supplementary checks	Recall and specificity of the first revision and the shipped rework on new annotated sentences, with source heterogeneity, NER-model, robustness and latency checks.
1	33 installed receiver probes	Request disposition on tested paths.
2	Matched head/full continuation; 16-case guard	Paired gains/harms and conditional acceptance.
3	Historical cost; installed V3 activation; telemetry audit, simulation and operation	Bounded cost, delivery/accounting and content-privacy/estimation limits.

adjudication. Cases sealed for an earlier semantic comparison were subsequently opened and reused for the product comparison and ablation. These remain development results on authored material, even when a later protocol was fixed prospectively.

A positive binary prediction requires nonzero sensitivity or a nonempty category set. Label-joint agreement requires exact matches for sensitivity, audience and category set; the *label-and-action joint* measure additionally requires an allowed action. We also measure allowed-action agreement separately. An action below the allowed set is under-restrictive, while one above it is over-restrictive. Receiver arrival provides a different observation: it establishes transmission rather than classification or policy agreement.

Keeping these outcomes separate prevents a single accuracy figure from concealing the mechanism of a change. Correct sensitivity with an incorrect audience fails exact label agreement; correct labels with an inadmissible action fail label-and-action agreement. Conversely, a different label tuple may still produce an allowed action. Binary recall and specificity deliberately discard these distinctions and answer only the declared presence question. We report paired gains and losses because the same net change can arise from preserving almost all earlier successes or replacing many of them with different successes. Appendix B defines the projections and paired measures precisely.

Final annotation target. The once-only final endpoint comprises 498 English rows from the Personally Identifiable Information Masking Benchmark (PIIMB), revision `4a13e9f`: 236 annotation-positive and 262 annotation-negative [21]. Source inputs, all six packages and scoring were frozen before opening, with no subsequent tuning, reselection or promotion. Sole-semantic execution evaluates the same nonzero-sensitivity-or-category projection against annotation presence. This target differs from contextual authorization and PIIMB's official character-masking/F2 task. The 464 observed source-group keys capture declared relationships, without proving template independence.

For contextual uncertainty, we use 2,000 paired whole-cluster draws with seed 42 and linear 95% percentile ranges, separately grouping by family and audience expression. Ablation expression groups were reconciled after collection. Final ranges use 2,000 paired source-group draws with the same seed and percentiles, retaining row weighting. These unadjusted ranges show sensitivity to known dependence; they establish neither crossed independence, template separation nor population coverage. Repeated calls and seeds likewise add no independent situations.

The grouping analyses address an identified dependence problem rather than repair the provenance of the targets. Resampling whole groups keeps their member rows together, but each analysis uses one declared grouping at a time. Family and audience-expression intervals cannot be combined as if their groupings were crossed-independent replications. Likewise, holding out rows by normalized text and known scenario group does not rule out common templates or semantic paraphrases. These limits apply even where source and model hashes establish that the same bytes were used.

B. Comparison and Retention Protocols

Historical and updated efficient, balanced and quality packages processed the same contextual cases through the product handler, with rules and policy fixed. Updated packages descend from frozen-encoder head fitting, joint encoder/pooler/head continuation and audience-supervised rehearsal; checkpoints were selected on validation alone before the sealed comparison. Historical packages used 64/96/128-token windows and category threshold 0.38, compared with 512 WordPieces and threshold 0.5 for updated packages. Their respective parameter counts were 0.019/0.037/0.054 and 4.785/9.596/17.494 million. These differences prevent attribution to pretraining or architecture alone, while repeatability passes add no cases. Rules and NER in this comparison and in the ablation below were first measured in the pre-change configuration of Section IV-C. Both were later rerun unchanged except for the revised detectors (Section V-D); the tables of Section V-A keep the historical measurements.

The comparison consequently asks whether a complete updated package behaves differently under the same product policy, rather than estimating a controlled effect of one training

stage. Its compact lineage identifies how the evaluated artifact was reached; it supplies no additional result catalogue. The updated package name must also be distinguished from the default installed selection: the earlier comparison evaluates frozen packages, whereas the later browser study verifies the standard V3 loading path. Sharing a family name does not make measurements from those environments interchangeable.

The ablation holds the updated balanced package, inputs and policy constant across full [2,3,4], no-rules [3,4] and no-NER [2,4] configurations. Execution checks distinguish requested layers from legitimate skips, allowing detector contributions on this workload to be examined without assuming that removal is safe elsewhere.

RQ2 holds the initial model and training exposure constant. Both head-only and full-model fits start from byte-identical official pretrained balanced encoder/pooler weights with zero task heads, rather than either RQ1 package. Training used Python 3.11, Torch 2.12/CUDA 13.0 and Transformers 4.57.6 on an RTX 5080. Seeds 42/43/44 received the same ordered 4,096 exposures over 64 steps, minimizing sensitivity and audience cross-entropy plus mean category binary cross-entropy with dropout disabled. AdamW used head/encoder rates $10^{-3}/2 \times 10^{-5}$, zero weight decay, norm-one clipping and microbatch 4/effective batch 64. Enabling encoder optimization in full-model training also changed the scope of joint-gradient clipping.

Excluding all declared training/guard groups and normalized texts removed 480 validation rows, leaving 1,080 reused provisional cases across 25 families: 570 sensitive and 510 clean. Each terminal candidate and the unchanged base underwent native in-process sole-semantic assessment; assessment scores selected no checkpoint. Sixteen separate guard cases required improved joint correctness and zero action harms against one derived target action per case, whereas the broader assessment accepted declared action sets. Guard acceptance and assessment improvement thus answer different questions. No candidate was published: effective-artifact mappings show what the rule would retain, conditional on the small guard and weak zero-head control.

The matched design reduces several sources of ambiguity: both treatments share the exact initial weights, ordered training exposures, head settings, objective and terminal assessment rule. It does not hold the effective optimization trajectory fixed, because allowing encoder gradients changes both the representation and joint clipping. The guard and broader assessment also serve different roles. A candidate's assessment gains describe behavior on the retained workload; its guard result determines eligibility under a separately declared rule. Neither result establishes what an installed client actually received. Appendix C records the resulting artifact and decision distinctions.

C. Detector Precision Protocol

The eighth evidence unit asks whether repaired rules, case-preserved input and the NER post-filter reduce false annotation-presence detections without an unacceptable loss of recall. It is a detector-layer study: each call explicitly requests the regex and NER layers, [2,3], and asserts that both executed successfully. The semantic layer is never requested, so the outcomes are neither semantic-only nor full-product evidence and are not comparable with Tables II or IV. The pre-change configuration is a clean export of the commit preceding the revision, with the original rules, lowercased detector input and unfiltered NER output used by the earlier units. The installed configuration is the first revision with all three controls of Section III-C, as it stood when the endpoint was scored; the later student-identifier rule and the labelled-identifier rework are not part of it. Single-control arms (case preservation alone, the post-filter on lowercased input and variants that ignore organizations) are development-only ablations reported in Appendix F; they are not held out.

Splits reuse existing material. The 502-row development set is divided by the parity of a hash of its source-group identifier into dev-TUNE (267 rows, 148 annotation-positive) and dev-CHECK (235 rows, 116 positive). Two synthetic sets complete the design: a personalized set of 2,000 rows (1,000 positive) and an OpenAI-generated set of 1,000 rows (500 positive). The post-filter rules were designed using only dev-TUNE and the personalized set, and were committed before any dev-CHECK or OpenAI score was computed. A subsequent review using author-written probe sentences produced a revised filter; because dev-CHECK and OpenAI had already been scored, the revised filter's results on them are post-review and no longer strictly held out. The data consulted while designing the regex repairs are not separately recorded, so the development and synthetic splits are treated as design material for them. The held-out set is the 498-row final endpoint of Section V-B (236 annotation-positive rows). It was scored once for both configurations after the installed defaults were frozen, as aggregate counts and timing only, with no row inspected and no rule or code changed in response to its rows; the student-identifier rule was diagnosed on the personalized set and added afterwards. The endpoint had already been opened for the semantic arms of unit 3 under its own protocol, so this is a one-shot detector-layer scoring rather than a new protected confirmation. The same frozen pair was later re-derived once more, storing per-row outcomes without text so that outcomes could be paired; its counts equal the first run exactly, and the endpoint is now consumed for this pair.

A prediction is positive when any finding has nonzero sensitivity or a nonempty category set, the projection of Appendix B. Because S1 findings are positive, downgrading an uncorroborated name from S2 to S1 changes no confusion cell; only dropping a finding does. We report true/false positives and negatives, recall, specificity and mean latency per prompt for regex, NER and their union. Labels are the source annotation-presence labels, not contextual authorization.

Paired uncertainty. A follow-up analysis re-derived per-row outcomes for the two frozen configurations on every split and checked that the confusion counts equalled those of Table V exactly. Because both configurations score the same rows, differences are tested on discordant rows only: an exact two-

sided McNemar test, separately for annotation-positive and annotation-negative rows, with rows that became correct called gained and rows that became incorrect called lost. Intervals for differences come from a paired percentile bootstrap (10,000 replicates, fixed seed) that resamples source documents on the development and held-out splits. The synthetic sets carry no template identifier, so their primary analysis resamples rows, which is likely anti-conservative for template-generated text; a conservative sensitivity analysis resamples the 20 personalized and 10 OpenAI scenario labels. No multiplicity correction is applied, so isolated small p-values are exploratory. The same tests are applied to the runtime action of the union (WARN or BLOCK, and BLOCK only), a different construct from annotation presence, and to costs weighted by false-negative to false-positive ratios of 2, 4 and 9.

D. Fresh Held-Out Protocol

The first held-out run reused an endpoint that had been opened before and, after the paired statistics, was consumed. Further detector-layer evidence therefore came from registered scoring of new sentences from the same benchmark (PIIMB, revision `4a13e9f` [21]), drawn so that no source group had appeared in the development or final sets, in any training or validation artifact, or in an earlier study. Hypotheses, margins, statistics and decision rules were registered before scoring, and the data, trees and scoring code were frozen in committed records before any new row was scored. Layers [2,3] were requested explicitly in every call and their execution asserted; no semantic layer, container or network was used.

The first registration (v2) tested the first revision N against the original detectors O on three sets. F1 holds 16,446 sentences (10,000 annotation-positive, 6,446 annotation-negative, 8,184 source clusters), F3 holds 2,000 long positive documents and F2 holds 6,000 unlabelled natural prompts from three public prompt collections. Its gating hypotheses were regex specificity (H1), union recall non-inferiority with a 1.0-point margin (H2) and union specificity non-inferiority (H2b); the same registration scored larger spaCy models and five default-off input canonicalizations against N (H3, H5, H6). After H2 failed, the owner chose to rework the rules that cost recall. A second registration (v3) tested the reworked K on F4, a new draw of 8,646 positive sentences from 5,745 source documents, disjoint from every earlier group including F1. Its primary hypotheses were union recall of K against O with a 1.0-point non-inferiority margin (H7) and BLOCK-level union recall with a 0.5-point margin (H8). Specificity guards ran on the F1 negatives (G1, K against O, margin 1.0; G2, K against N, margin 2.0), which O and N had already been scored on and whose published results informed the rework, so a pass there means not contradicted rather than freshly confirmed. F4 contains positives only.

Intervals are paired percentile bootstrap intervals (10,000 replicates, seed 20261024) that resample whole source-group clusters; a non-inferiority test uses the lower limit of the two-sided 95% interval, which is the one-sided 97.5% bound. A hypothesis passes only if the whole bound lies on the favourable side of its margin. An interval that straddles the margin is inconclusive, and the pre-registered consequence of a failure to pass is a stated trade-off, not promotion. Each delta is reported with rows gained and lost and the exact McNemar value. Analyses fixed before scoring are called registered. Analyses produced afterwards from the same outputs (layer swaps, per-rule tables, length and category strata, critic recomputations) are exploratory and cannot satisfy or rescue a registered verdict. F2 results are flag rates only: the prompts are unlabelled, and the human adjudication of a flagged sample has not been done.

E. Enforcement, Cost and Telemetry Protocols

Thirty-three installed Linux Chromium probes exercised the built extension, standard native installer, supervisor, bridge and historical balanced detector with the pre-change rules and NER. A local Hypertext Transfer Protocol Secure (HTTPS) receiver checked original bodies and late sends during forwarding, blocking, faults, cancellation and recovery. Product and sole-semantic probes outside the resource windows asserted actual layer execution. These synthetic receiver observations do not establish authenticated-provider or cross-platform compatibility.

Historical cost measurements used three fixed synthetic cases, one per action, over fetch and XHR. Two fresh profiles per transport/action cell yielded twelve profiles, each completing a first decision, five warmups and thirty measured rules/NER requests, for 360 measured decisions. Page latency ran from analysis request to result, excluding receiver response; nearest-rank percentiles pool only two profile clusters per cell. Resource windows began at browser launch and ended after request/readiness checks, before teardown. Whole-container central processing unit (CPU) time and sampled charged memory include infrastructure, cache, driver, receiver and observer. Charged memory differs from process resident set size (RSS) and proportional set size (PSS) [22]. Sampling requested 100-ms intervals and recorded actual gaps. The historical V1 study included neither semantics, an unprotected baseline nor per-role resource attribution.

The V3 study used two fresh Linux Chromium profiles per transport and the synthetic prompt “Please explain the steps in this process.” Each profile retained its normal launcher, supervisor, bridge and model service across detectors-off, rules/NER, semantic and full arms, with order reversed in the second replicate. Five warmups and thirty measurements per arm produced 80 warmups and 480 timed requests. Because every timed request forwarded, request-to-receiver completion could be compared through within-profile median subtraction. The off baseline retained interception and infrastructure. Exact layers and packages were asserted on Torch 2.11 CPU with four threads, Python 3.12.3 and Chromium 153.0.8010.12; telemetry was disabled. Ten untimed post-cost probes checked the three V3 tiers, older V1 balanced and return to default over both transports.

The telemetry audit enumerated exact rational kernels on a fingerprinted Windows CPython 3.11.9 build, enclosing loga-

rithms with outward 100-digit arithmetic under independent uniform operating-system (OS) bits. Budgets 0.5, 1 and 2 cover the declared settings. Estimator simulation crossed these budgets with $n = 8, 80, 800, 8000$, all five field domains, and balanced-integer, point-mass and minimum-one 1%-target histograms. Each of the resulting 180 prespecified cells had 10,000 repetitions with seed 20261014. NumPy 1.26.4/PCG64 sampled binary64 approximations of finite probabilities; the collector applied nominal inversion and independent $[0, n]$ clipping. Large n pools installation-days, still limited to eight default reservations each. The estimand is a fixed contributor histogram, excluding population-sampling variation, and no deployment-usefulness threshold was specified.

Ledger checks examined persistence, concurrent admission and failure accounting. A separate controlled Windows host study exercised the actual reporter, randomizer and ledger with a separate SQLite-backed collector, using generated analysis-shaped fixtures without a detector. Twenty-three ledger fixtures covered caps, reopening, ordered inputs, outage/recovery and malformed packets. Synchronous report-call timing excluded background delivery and was kept separate from Linux browser timing. Neither study involved real-user traffic.

We therefore use the operational fixtures to test report delivery and budget accounting, and the simulation to study estimator behavior conditional on known contributors. They cannot be pooled into a user study. The browser experiment likewise evaluates an installed local path to a synthetic receiver; its instrumentation and resident services are part of the observation boundary. Appendix D distinguishes the timing endpoints, and Appendix E explains why reservation counts, received reports and potential users require different denominators.

Artifact availability. Tracked recipes in `evaluation/` and dated records in `paper/research/` identify configurations through source, model and data fingerprints and environment receipts. They distinguish product/ablation dependencies, including historical uncommitted sources, from matched continuation, final semantics, historical and standard-V3 client captures, telemetry, and the detector-precision unit with its follow-up (`detector-precision-20261024.md` and `detector-precision-followup-20261024.md`). Some pinned checkpoints and raw run artifacts remain local and ignored. Exact reruns therefore require more than a Git checkout, and public distribution of those assets has not been established.

V. RQ1: Detection and Enforcement

A. Contextual Policy Agreement

Updated packages more often matched both the authored labels and an allowed action. Label-and-action joint agreement rose by 27.37–41.86 percentage points to 693–879 of 1,560 cases (Table II), with no failed calls or layer errors in any of the six runs. Efficient, balanced and quality gained 484/573/671 previously incorrect joint outcomes while losing 57/16/18 previously correct ones. Allowed-action gains were 389/389/348, against losses of 3/6/0. Quality thus preserved every previous allowed-action match while losing eighteen joint matches, illustrating that an intervention can be retained even when its classification changes. These gains concern reused provisional targets; no package was promoted.

TABLE II
Full-product comparison on the same provisional contextual cases, shown as historical/updated counts. Labels+action is label-and-action joint agreement and Action is allowed-action agreement, both out of 1,560. Sensitive and Clean count correct classifications out of 810 and 750. All arms completed without errors. Historical pre-change detectors; Table VII reports the revised detectors.

Package	Labels+action	Action	Sensitive	Clean
Efficient	266/693	1029/1415	544/808	494/629
Balanced	160/717	1026/1409	771/802	318/631
Quality	226/879	1073/1421	598/810	468/641

Execution records identify a shared source of unnecessary restriction. All updated packages over-restricted the same 139 cases: rules blocked 109 clean controls before later detectors could run and warned on thirty low-risk controls. Rules ran on every case, ending the cascade on 289 and leaving NER and semantics to process 1,271. A better downstream semantic model cannot undo these earlier decisions. These rule-block counts are historical measurements of the pre-change rules; Section V-D reports the rerun with the revised detectors, under which the shared figure falls to 108. In the historical configuration NER produced zero converted PriVoke findings after supported-label filtering, although this does not mean that the recognizer was inactive or returned no raw entities; that statement does not hold for the revised detectors.

The fixed-package ablation examines this explanation by comparing agreement with under- and over-restriction (Table III). Removing rules corrected all 139 shared unnecessary restrictions but lost eighteen previously correct actions. Its net gain of 121 actions therefore came at the cost of missed protection under the provisional policy. Paired action-rate gains ranged from 3.46–12.70 percentage points by family and 5.96–9.23 by audience expression. These separate descriptive ranges support the direction of the tradeoff without establishing that its losses are acceptable. Historically, No-NER outputs equalled the full product, consistent with zero converted findings on this fixture. Verified execution supports that observation, but neither ablation establishes that a detector is unnecessary elsewhere. Table III is a historical measurement with the pre-change rules and NER; the rerun with the revised detectors in Section V-D qualifies both the equality and the 139.

B. Agreement on a Separate Annotation Target

The final endpoint asked whether the packages distinguished positive from negative rows under a different annotation target. All 2,988 semantic-only calls completed without errors and verified exclusive execution of the requested layer. Updated recall reached 80.51–93.64%, but specificity remained at only 13.36–21.76% (Table IV). Efficient and balanced lost

TABLE III

BALANCED-PACKAGE ABLATION ACROSS THE SAME 1,560 PROVISIONAL CASES, WITH REQUESTED LAYERS IN BRACKETS. ACTION AND JOINT COUNT ALLOWED-ACTION AND LABEL-AND-ACTION JOINT AGREEMENT; UNDER AND OVER COUNT ACTIONS BELOW AND ABOVE THE ALLOWED SET. HISTORICAL PRE-CHANGE DETECTORS; TABLE VIII REPORTS THE REVISED DETECTORS.

Configuration	Action	Joint	Under	Over
Full [2,3,4]	1409	717	12	139
No rules [3,4]	1530	972	30	0
No NER [2,4]	1409	717	12	139

TABLE IV

FINAL SEMANTIC ANNOTATION-PRESENCE RESULTS FOR 236 POSITIVE AND 262 NEGATIVE ROWS. HISTORICAL/UPDATED COUNTS SHOW TRUE/FALSE POSITIVES (TP/FP) AND TRUE/FALSE NEGATIVES (TN/FN). ALL SIX ARMS COMPLETED WITHOUT ERRORS.

Package	TP	TN	FP	FN
Efficient	140/218	61/45	201/217	96/18
Balanced	141/221	65/35	197/227	95/15
Quality	167/190	51/57	211/205	69/46

specificity relative to their historical versions, while quality's small increase remained uncertain. Paired net-accuracy gain ranges from frozen source-group resampling were nevertheless positive: 7.41–17.37, 5.35–15.03 and 1.02–10.71 percentage points, respectively. Recognizing additional positives improved average accuracy despite poor discrimination of annotation-negative rows.

This endpoint complements rather than reproduces the contextual comparison. It assesses source annotation presence through semantics alone, whereas the product comparison assesses provisional contextual labels and actions through a cascade. The difference between their results is therefore not a controlled estimate of distribution shift or detector contribution. Low final specificity still demonstrates poor discrimination under the declared projection, but neither endpoint independently establishes whether disclosure is authorized.

C. Detector-Layer Precision Revision

The preceding results attribute much of the unnecessary restriction to rules and the weak specificity of annotation presence to detectors that fire without personal context. The precision revision of Section III-C targets that behavior at the detector layers. Table V compares the pre-change and installed configurations for regex, NER and their union on the splits of Section IV-C. These are detector-layer measurements: they exclude the semantic layer, policy mapping and request enforcement, and they are not comparable with the product or semantic-only tables above.

On the held-out final endpoint, the regex layer's specificity point estimate moved from 0.985 to 0.996 (false positives 4 to 1) while its recall moved from 0.653 to 0.640 (true positives 154 to 151). The design and development splits show the same direction at similar size: dev specificity 0.971 to 0.983 and recall 0.652 to 0.640. The synthetic sets show larger

TABLE V

DETECTOR-LAYER ANNOTATION-PRESENCE COUNTS (TP/TN/FP/FN), SHOWN AS PRE-CHANGE/INSTALLED. REGEX AND NER WERE REQUESTED EXPLICITLY ([2,3]); UNION IS POSITIVE WHEN EITHER LAYER REPORTS. THE DEVELOPMENT SET HAS 502 ROWS (264 POSITIVE), DEV-TUNE 267 (148), DEV-CHECK 235 (116), PERSONALIZED 2,000 (1,000), OPENAI 1,000 (500) AND THE HELD-OUT FINAL ENDPOINT 498 (236). †POST-REVIEW SET.

Split	Layer	TP	TN	FP	FN
dev (502)	Regex	172/169	231/234	7/4	92/95
	NER	55/65	232/230	6/8	209/199
	Union	193/201	225/226	13/12	71/63
dev-TUNE	Regex	96/95	115/117	4/2	52/53
	NER	35/41	117/115	2/4	113/107
	Union	113/118	113/113	6/6	35/30
dev-CHECK†	Regex	76/74	116/117	3/2	40/42
	NER	20/24	115/115	4/4	96/92
	Union	80/83	112/113	7/6	36/33
Personalized	Regex	1000/1000	977/1000	23/0	0/0
	NER	800/695	1000/1000	0/0	200/305
	Union	1000/1000	977/1000	23/0	0/0
OpenAI†	Regex	460/448	346/466	154/34	40/52
	NER	145/85	492/497	8/3	355/415
	Union	471/459	343/463	157/37	29/41
Held-out	Regex	154/151	258/261	4/1	82/85
	NER	56/50	255/253	7/9	180/186
	Union	181/180	251/252	11/10	55/56

specificity gains: regex false positives fell from 23 to 0 on the personalized set and from 154 to 34 on the OpenAI set, where regex recall fell from 0.920 to 0.896. Because the rules were developed with these sets in view, the synthetic gains are optimistic, and the held-out difference is a net difference of three rows.

Paired tests on the same rows (Table VI) show that the held-out regex shifts are not statistically significant. Three rows became correct and none incorrect for specificity, and none became correct and three incorrect for recall; the exact McNemar p-value is 0.25 for each, and the 95% intervals for the differences, [0.000, 0.026] and [−0.029, 0.000], reach zero. The development split shows the same three-against-zero counts. The point estimates above are therefore consistent in direction but compatible with no change. The synthetic-set regex changes are significant: personalized false positives fell on 23 rows with none added ($p < 0.001$, although an interval that resamples its 20 scenario labels touches zero, [0.000, 0.073]), and the OpenAI set gained specificity on 121 rows against 1 ($p < 0.001$; scenario-label interval [0.174, 0.329]) while losing recall on 12 rows against none ($p < 0.001$).

The union shows no detectable change on held-out data: recall 0.767 to 0.763 (181 to 180 true positives; 13 rows gained and 14 lost, $p = 1.0$) and specificity 0.958 to 0.962 (11 to 10 false positives; 7 gained and 6 lost, $p = 1.0$). This supports no detectable change, not equivalence: the intervals for the differences, [−0.047, 0.039] for recall and [−0.023, 0.031] for specificity, remain compatible with shifts of three to five

TABLE VI

PAIRED DETECTOR-LAYER DIFFERENCES, NEW CONFIGURATION AGAINST PRE-CHANGE. GAINED AND LOST COUNT ROWS THAT BECAME CORRECT OR INCORRECT FOR THE STATED CLASS (POSITIVE ROWS FOR RECALL, NEGATIVE ROWS FOR SPECIFICITY); p IS THE EXACT TWO-SIDED MCNEMAR VALUE ON THOSE DISCORDANT ROWS. NO MULTIPLICITY CORRECTION IS APPLIED.

Split	Layer	Measure	Gained	Lost	p
Held-out	Regex	Specificity	3	0	0.25
Held-out	Regex	Recall	0	3	0.25
Held-out	Union	Specificity	7	6	1.0
Held-out	Union	Recall	13	14	1.0
Held-out	NER	Specificity	4	6	0.75
Held-out	NER	Recall	23	29	0.49
Dev	Union	Recall	12	4	0.077
Dev	NER	Recall	26	16	0.16
Personalized	Regex	Specificity	23	0	<0.001
Personalized	NER	Recall	52	157	<0.001
OpenAI	Regex	Specificity	121	1	<0.001
OpenAI	Regex	Recall	0	12	<0.001
OpenAI	NER	Recall	11	71	<0.001

points either way. The development-set rise in union recall from 0.731 to 0.761 (12 rows gained, 4 lost) is not significant by the exact test ($p = 0.077$) and did not reproduce on the held-out endpoint. The NER layer alone did not improve there: false positives rose from 7 to 9 and recall fell from 0.237 to 0.212. It also lost recall on both synthetic sets (personalized 0.800 to 0.695, OpenAI 0.290 to 0.170), and gained recall on the development set (0.208 to 0.246, with specificity 0.975 to 0.966) and on dev-CHECK, where its false positives stayed at four while recall rose from 0.172 to 0.207. The NER differences on real text are inside the noise and have opposite signs (development 26 rows gained against 16 lost, $p = 0.16$; held-out 23 against 29, $p = 0.49$), whereas the synthetic recall losses are significant (personalized 52 against 157, OpenAI 11 against 71, $p < 0.001$ each). A net NER benefit is therefore not established. The NER changes are installed on design and development evidence, with an off switch, and the union result is dominated by the regex layer. The direction of the trade is regex specificity for a small recall cost, but on real text neither side is statistically established; only the synthetic-set changes are.

Action-level analysis of the union separates a change in annotation presence from a change in the runtime action. On the held-out endpoint no WARN-or-BLOCK difference is distinguishable from zero (recall 0.606 to 0.602, specificity 0.977 to 0.973), BLOCK-only recall moves from 0.398 to 0.386 (none gained, three lost, $p = 0.25$), and no difference in cost-weighted error at false-negative to false-positive ratios of 2, 4 and 9 is distinguishable from zero. The largest action-level change appeared on the personalized set: BLOCK-only recall fell from 1.000 to 0.928 because 72 positives moved from BLOCK to WARN (72 against 0, $p < 0.001$). Annotation presence cannot show this, since recall stays at 1.000. The cause was removal of a regex child category that had matched records of adult students by accident. The student-identifier rule added afterwards (Section III-C) restored BLOCK on those rows without new false positives, but it rests on a single template, so the mechanism is confirmed and its generalization is not. The tables of this subsection and of the full-product rerun do not include the rule; the fresh-data evaluation of Section V-E does.

TABLE VII

FULL-PRODUCT RERUN OF THE CONTEXTUAL COMPARISON, SHOWN AS HISTORICAL PRE-CHANGE/REVISED-DETECTOR COUNTS OUT OF 1,560. JOINT AND ACTION ARE LABEL-AND-ACTION AND ALLOWED-ACTION AGREEMENT; OVER AND UNDER COUNT ACTIONS ABOVE AND BELOW THE ALLOWED SET.

Package	Joint	Action	Over	Under
V1 efficient	266/280	1029/1060	297/267	234/233
V1 balanced	160/155	1026/1037	405/396	129/127
V1 quality	226/237	1073/1117	251/218	236/225
V3 efficient	693/702	1415/1443	139/111	6/6
V3 balanced	717/720	1409/1438	139/110	12/12
V3 quality	879/876	1421/1451	139/109	0/0

Mean detector latency per prompt, from one sequential pass, was 44.3 to 52.4 ms for the pre-change configuration and 46.1 to 51.5 ms for the installed one across splits, with no meaningful difference. Detector-layer results do not by themselves establish the effect on contextual actions; the next subsection reports a full-product rerun of the comparison and ablation. The installed probes and the cost increments reported later were not rerun and remain pre-change measurements. The comparison also shares the limits listed in Section VIII-B: one small spaCy model, annotation-presence labels with their noise, S1 findings counted as positive, and partly reused design material.

D. Full-Product Rerun with the Revised Detectors

The comparison and ablation of Section V-A were measured with the pre-change detectors. They were rerun as full-product evidence (rules, NER and semantics, actual cascade execution recorded), not as detector-layer or semantic-only results. The rerun used the same 1,560 cases, six packages, policy, rule-first ordering and one first pass per cell, and compared a clean export of the pre-change commit with the revised detectors without the student-identifier rule. The pre-change arm reproduced every historical outcome and action on all 1,560 cases for all six packages and seven layer sets, with no discordant pair, so differences arise from the detectors rather than the harness. The semantic layer is unchanged by construction, and its sole-semantic outcomes were identical in both arms. All 40,560 scored requests completed without errors. Intervals resample the 35 scenario families.

The revised detectors reduced over-restriction in every package, and the 139 unnecessary restrictions shared by the three V3 tiers fell to 108 (112 in union); 103 of the 108 lie among the original 139. The residual shows that the over-enforcement was largely, but not entirely, a rule artefact. Allowed-action agreement rose for every package, for example V3 balanced from 1,409 to 1,438 (36 cases improved, 7

TABLE VIII
FULL-PRODUCT RERUN OF THE BALANCED-PACKAGE ABLATION, SHOWN AS HISTORICAL PRE-CHANGE/REVISED-DETECTOR COUNTS OUT OF 1,560, WITH REQUESTED LAYERS IN BRACKETS.

Configuration	Action	Joint	Under	Over
Full [2,3,4]	1409/1438	717/720	12/12	139/110
No rules [3,4]	1530/1523	972/942	30/30	0/7
No NER [2,4]	1409/1445	717/745	12/12	139/103

regressed, $p < 0.001$), and binary false positives for that package fell from 119 to 95 with false negatives unchanged at 8; no V3 tier gained a binary miss. The mechanism for V3 balanced was 36 cases moving from BLOCK to ALLOW as rule false positives disappeared (early rule BLOCKs fell from 289 to 253), against 7 clean cases moving from ALLOW to WARN because NER now emitted findings. Joint agreement did not change significantly for V3 (p from 0.26 to 0.81) and fell by five for V1 balanced (2 improved, 7 regressed, $p = 0.18$). Two effects cancelled: the rule repairs alone added 28 joint-correct cases without a loss, while the new NER findings altered the label set on 25 cases, each in favour of removing NER. The updated V3 packages remained ahead of historical V1 in every tier, with allowed-action gains of 388 against 5 losses for efficient, 413 against 12 for balanced and 337 against 3 for quality, compared with 389/3, 389/6 and 348/0 historically.

The central ablation result persists but shrinks. Removing rules versus the full product corrected 139 actions and lost 18 historically (net 121), and with the revised detectors corrected 103 and lost 18 (net 85, $p < 10^{-14}$; family-resampled interval for the action-rate gain 2.4 to 8.9 percentage points, against 3.5 to 12.7 historically). The lost actions are again 18, and under-restriction again rises from 12 to 30, so the tradeoff between unnecessary restriction and missed protection remains. The revised detectors also change the NER findings. Historically NER produced no converted findings on this fixture and removing it left every outcome unchanged. With case-preserved input and the post-filter it converts 48 findings, all identity names: alone it flags 36 of the 810 sensitive-labelled cases and 12 of the 750 clean cases. No-NER now equals the full product on 1,526 of 1,560 cases, and removing NER improves 7 action outcomes and 25 joint outcomes with no regression. On this fixture the NER layer is therefore a net negative at the label level, which does not establish its value or harm elsewhere.

These are full-product measurements on the same assistant-provisional labels and correlated families as before. The detector changes were tuned on development and personalized sets rather than on these cases, but this partition has now been scored twice, so the rerun is not a fresh confirmation. The ablation uses one balanced package, the intervals use the family grouping only, and neither the student-identifier rule nor the later rework (K) is part of the rerun: the shipped configuration has not been run through the full product.

E. Registered Fresh Held-Out Evaluation

The preceding detector-layer results were consumed or post-review. Registered scoring on new sentences (Section IV-D) tested the first revision N and then the reworked K against the original detectors O. Table IX gives the registered cells and the context cells on the same rows. These are detector-layer results under annotation presence and the runtime action of the union, not semantic-only or full-product evidence.

The first revision failed its recall test. N raised regex specificity by 1.75 points and union specificity by 3.32 points on F1, but its union recall fell 1.37 points against O, and the whole interval lies below zero as well as below the -1.0 margin ($p = 2 \times 10^{-6}$). The loss is concentrated at the BLOCK level: 190 positives moved from BLOCK to a lesser action and none moved the other way, all through the regex layer. By the registered rule N was not promotable; N was already the installed default, and the registration's stated consequence was the trade-off itself, not a revert. An exploratory exact layer swap placed the recall loss in the regex layer: keeping the old regex layer with the new NER layer gave $+0.08$ points of union recall (-0.43 to $+0.61$), whereas the new regex layer with the old NER layer gave -1.49. The rules behind the 146 positives lost through the regex layer were mainly the weakly shaped licence recognizer, IBAN and its validator, concentrated-sensitive-data, the card and social-security recognizers, and the child and family cues; the child and family hits fell mostly on text the annotation does not mark. This attribution is post hoc and uses no new data. It prompted the owner's decision to rework the regex rules that cost recall.

The rework recovered part of the loss and not all of it. On F4, K lost 1.05 points of union recall against O (342 gained, 433 lost). The estimate sits 0.05 points beyond the margin, the interval straddles it, and H7 is inconclusive. Relative to the margin that means the data neither show a loss within 1.0 point nor show that the loss exceeds it; relative to zero, the upper limit -0.41 shows K is significantly worse than O on union recall (exact McNemar $p = 0.0012$). BLOCK-level recall is level (H8). Against N, K dominates on every recall rule: 69 union gains and no losses, and 135 BLOCK gains and none lost. K restores about 43% of N's union-recall loss on F4 (-1.85 for N, -1.05 for K) and all of its BLOCK-level loss (-1.55 for N, $+0.01$ for K). The 5 clean F1 rows K adds against N, all from the generic labelled-identifier rule, are the cost of the new rules, and the G2 pass is a second look. By the registered rule K was not promoted, because H7 did not pass. It is nonetheless the shipped configuration: the owner decided on 25 October 2026 to ship the rework, and an independent critic reviewing the registered cells advised that if the first revision shipped, K was the better version. The critic's summary concluded that K's recall loss against O is real, that BLOCK-level recall is equal, and that reverting to O would buy back about one point of union recall, mostly through O's noisy NER, at about three points of union specificity.

The result depends on the source. Pooled F4 averages gains

TABLE IX

REGISTERED FRESH HELD-OUT RESULTS, DETECTOR LAYERS ONLY (REGEX OR NER, POSITIVE RULE UNLESS THE ACTION IS NAMED). DELTA IS CANDIDATE MINUS REFERENCE IN PERCENTAGE POINTS WITH A PAIRED GROUPED 95% INTERVAL; GAINED/LOST COUNT ROWS THAT BECAME CORRECT OR INCORRECT. F1 HOLDS 10,000 POSITIVE AND 6,446 NEGATIVE SENTENCES AND F4 HOLDS 8,646 POSITIVE SENTENCES. MARGIN IS THE REGISTERED NON-INFERIORITY MARGIN. †SECOND LOOK AT NEGATIVES SCORED BEFORE K WAS FROZEN, NOT A FRESH TEST. ‡REGISTERED SECONDARY OR CONTEXT CELL, NOT A HYPOTHESIS.

Cell	Delta [95% CI]	Gained	Lost	Verdict
H1 regex specificity, N−O, F1 negatives	+1.75 [+1.42, +2.11]	114	1	pass
H2 union recall, N−O, F1 positives (margin −1.0)	−1.37 [−1.95, −0.79]	342	479	**fail**
H2b union specificity, N−O, F1 negatives	+3.32 [+2.60, +4.07]	385	171	pass
Union BLOCK recall, N−O, F1 positives‡	−1.90 [−2.18, −1.63]	0	190	loss
H7 union recall, K−O, F4 (margin −1.0)	−1.05 [−1.70, −0.41]	342	433	inconclusive
H8 union BLOCK recall, K−O, F4 (margin −0.5)	+0.01 [−0.33, +0.35]	104	103	pass
Union recall, K−N, F4‡	+0.80 [+0.62, +0.99]	69	0	context
Union BLOCK recall, K−N, F4‡	+1.56 [+1.29, +1.84]	135	0	context
G1 union specificity, K−O, F1 negatives† (margin −1.0)	+3.24 [+2.53, +3.98]	385	176	not contradicted
G2 union specificity, K−N, F1 negatives† (margin −2.0)	−0.08 [−0.15, −0.02]	0	5	not contradicted

with a large loss in one source (Table XV, Appendix G). In the Privy sentences K lost 8.00 points of union recall against O (interval −9.57 to −6.41), the same loss as N (−8.04), so K's rework leaves it unchanged: K differs from N on one Privy row. The F1 draw showed the same Privy loss for N (−6.57) and a larger loss in Nemotron-PII (−4.86, against −2.35 on F4). Outside Privy, K gains 1.25 points (+0.59 to +1.92). F4 is Privy-heavy (25% of rows) and Nemotron-light (8%), the mix the registration flagged as unfavourable; reweighting to the F1 mix gives −0.43 (−1.06 to +0.24), whereas equal source weights give −1.45. Exploratory analysis of the Privy losses found that 164 of 241 lost rows had old NER spans that touch no annotated span, and about 77 had a span touching an annotated one; on F1 the old union also flagged 48% of Privy negatives against 29% for N and K. Much of the old Privy recall is therefore indiscriminate flagging, but because the benchmark's annotation is incomplete by construction, unannotated personal data in those rows cannot be excluded, so the loss may be partly false-positive reduction and partly real misses. Losses concentrate in long sentences (203 or more characters, −3.21 points) and in location, organization and person annotations, categories carried mainly by the NER layer, while K gains in given-name, surname, street and building-number annotations. The NER layer shared by N and K lost 4.58 points of NER-only recall against O on F4.

Flag rates on natural prompts. Of 6,000 unlabelled prompts, O flagged 24.4% (1,465), whereas N and K each flagged 16.7% (1,001), and K flagged exactly the prompts N did. BLOCK fell from 130 prompts to 65. These are flag rates, not error rates; whether flagged prompts contain personal data awaits human adjudication, so no specificity claim is made on natural prompts. The generic labelled-identifier rule fires at S2 on technical text such as a user-identifier key and value, and its qualifier vocabulary mirrors the labels of the development spans, so its gains may not transfer to text that does not use such labels.

Spans. Counting a prompt as detected when any rule fires overstates masking. On F1, N's union detects 80.92% of positive sentences by that definition, yet fully masks every annotated span in only 37.16% of them (37.31% for O). Span-level scoring on the 502 development sentences showed the same gap (union any-hit recall 0.761, every mention masked in 107 of 264 prompts, 0.405, and 0.625 of gold characters covered). Span measures also moved differently from the presence measures: on F1, N's union overlap recall rose 1.19 points and its over-masking fell by 12.75 characters per thousand, whereas on F4, K raised gold-character coverage by 5.00 points over O but masked 10.82 more characters per thousand outside annotated spans than N. Letting NER extend masking only where the regex layer already fired recovers part of the NER span gain without creating a positive.

F. Supplementary Detector Checks

Five further tests ran in parallel with the registrations. None changes the shipped configuration.

NER models. Two larger spaCy models and a GLiNER-based personal-data model were compared with the shipped small model on dev-CHECK and the OpenAI set, with the GLiNER label preset and threshold frozen on the design splits first. No union difference among the candidates with the post-filter on was statistically reliable (smallest exact $p = 0.125$). The larger spaCy model had the most consistent gains on those post-review splits (dev-CHECK 5 true positives gained and 1 lost, 4 false positives removed; OpenAI 1 gained, 3 removed) and was neutral on the design splits. The GLiNER model lost NER recall (dev-CHECK −0.103, $p = 0.008$), left the union level or lower, and documents no training data, so it was dropped. The transformer spaCy model is a comparator at roughly six times the NER-layer latency. On the fresh F1 set the larger model raised union specificity over N by 1.71 points (1.17 to 2.24) and union recall by 1.86, but it lowered specificity in the Privy sentences (union −4.34, NER-only −7.59), so the pooled gain depends on the source mix. It also adds about 550 MB on disk and 762 MB resident. None of the evaluated models changed a BLOCK decision on the

post-review splits, because their entities are at most S2. The owner held the larger model, and the shipped NER model is unchanged.

NER as amplifier. A study on the development splits asked whether NER is better used only to corroborate or extend regex findings. Corroboration-only policies never beat the shipped union on the design splits, and amplifier-only policies cannot create a positive by construction, so they surrender NER's recall gain (negative result for NER as an amplifier-only detector). Modest independent recall gains from the shipped NER layer remain.

Robustness probe and fixes. A perturbation probe applied seeded transformations to the development, personalized and OpenAI sets. Digit obfuscation caused the largest recall gap: splitting digits with dashes or spelling them out cost the revised union about 20 to 29 points on development and OpenAI text, and on OpenAI the precision repairs made it worse than O (regex recall fell 34.4 points against 15.2 under dashes). Zero-width characters cost 11 points of development union recall and inflated NER false positives; lowercase and uppercase text reduced NER recall; digit spacing, full-width digits and bracketed at-signs were already neutralized. Four default-off canonicalizations were then built and scored on F1's 58,899 perturbed items: stripping zero-width characters (+9.11 points of union recall), rejoining split or spelled digits (+17.82), spelled e-mail separators (+30.90) and a caseless gate on NER (+1.30), together +11.77, all passing their registered tests. These passes are largely circular. Stripping restores the unperturbed result on every item by construction (zero mismatches), the digit gain comes from the dash and spelled transforms, the others having been handled already, the caseless gain is entirely in uppercase text, and the no-harm test passes because no clean sentence changed outcome. The tests show each fix works on its own transform and say nothing about how often such inputs occur. The documentation prerequisite registered for promotion was unmet at the frozen commit; the owner waived it, and the fixes remain off and outside K.

Latency. In a host-process, detector-layer timing (1,000 prompts, three repeats, fresh worker per run), the combined regex and NER call took a median 27.3 ms per prompt for O, 27.5 for N and 27.76 for K against 27.77 for N in the same session. The robustness flags and the larger spaCy model changed latency by less than the run-to-run spread of about 1 ms, though the larger model raised resident memory from 387 to 1,149 MB. The transformer model added 50.1 ms at the median and 2.4 GB of memory, a 211% increase in mean latency against O, which fails the registered 20% criterion. This is not an installed-browser measurement.

Annotation study. A blinded human double-annotation protocol for the contextual targets, with sampling, agreement statistics and a check of the provisional targets, has been written and its scripts tested on synthetic inputs. It has not been run, and no contextual claim depends on it.

G. Does a Decision Control Transmission?

Across thirty-three correctness probes, the installed-browser capture connected decisions to eighteen unchanged-body sends and fifteen zero-send outcomes. Correlated runtime and receiver records, including late-arrival checks, established these outcomes without a decision broker. Healthy full-pipeline and sole-semantic probes also verified execution, linking request disposition to the detector path actually exercised.

Fault probes exposed the difference between keeping a request path available and preserving its protection. When the parameter stream was interrupted, semantic-only requests were blocked, but configured full-mode fallback ran rules and NER and forwarded the request. Restarting restored full execution; disabling protection or all layers forwarded by design. The fallback thus maintained reduced analysis without demonstrating equivalent semantic protection. These completed observations were kept separate from nine earlier failed captures.

Answer to RQ1. PriVoke improved agreement with provisional contextual policy and enforced its declared request contract on the tested installed paths. Action errors, unchanged-content forwarding under WARN and weak specificity on the separate annotation-presence task constrain the broader effectiveness claim. A later detector-layer revision targets the rule and NER false detections behind part of that restriction. A full-product rerun measured the first revision's effect on actions, but the shipped rework was evaluated only at the detector layers, where registered fresh tests found a real union-recall cost against the original detectors alongside a specificity gain. The results do not yet establish independent contextual accuracy, provider compatibility, prevention for every sensitive prompt or practical usability.

VI. RQ2: Updates and Previously Protected Behavior

A. Improvement Subject to Retention

A higher score does not necessarily make a candidate suitable for replacement. Whereas RQ1 changes capacity, tokenizer, threshold and supervision together, RQ2 fixes the base and exposure before enabling encoder optimization. This matched comparison separates the observed learning gain from the decision to permit publication.

B. Matched Continuation and Guard Decisions

All six 64-step fits completed with finite parameters. HEADS preserved the encoder tensors, while FULL changed 39 and produced nonzero gradients in later steps. The six candidates and unchanged base completed 7,560 error-free native in-process semantic assessments, each covering the same 1,080 situations. These repeated assessments provide neither additional independent cases nor measurements of deployed network execution.

FULL improved joint and action agreement in every seed (Table X). For seeds 42/43/44, joint gains and losses were 79/14, 69/26 and 90/20; action gains and losses were 123/38, 124/41 and 110/19. Under-enforcement distance increased on 33/37/15 cases and over-enforcement distance on 17/13/17.

TABLE X
MATCHED SEMANTIC-ONLY CONTINUATION ON 1,080 REUSED PROVISIONAL CASES, SHOWN AS HEAD-ONLY/FULL-MODEL COUNTS. LABELS+ACTION DENOTES LABEL-AND-ACTION JOINT AGREEMENT. THE UNCHANGED ZERO-HEAD BASE HAS ZERO JOINT MATCHES AND 480 ALLOWED-ACTION MATCHES. GUARD HARMS ARE RELATIVE TO THAT BASE ON SIXTEEN SEPARATE CASES.

Seed	Labels+action	Allowed action	Guard harms
42	192/257	908/993	0/1
43	204/247	906/989	1/1
44	198/268	912/1003	0/1

These directional counts do not form a mutually exclusive harm union and cannot be summed as distinct harmed cases. Aggregate improvement therefore coexisted with lost protection on reused cases, without establishing how frequently either would occur in a population.

Each FULL candidate introduced one action harm relative to the unchanged base on the sixteen guard cases and was rejected. HEADS42 and HEADS44 passed, whereas HEADS43 failed. Mapping a rejected candidate back to the base forfeits its assessment gains, but these mappings remained hypothetical: no candidate in this experiment was published or promoted. Passing a small guard cannot rule out other harms or compensate for the weak zero-head baseline.

Answer to RQ2. Joint continuation of the pretrained encoder and task heads improved aggregate agreement over matched head-only training in all three seeds. Nevertheless, every FULL candidate, together with head-only seed 43, failed retention. The experiment demonstrates bounded learning gains and an enforceable guard, but those gains did not meet the retention requirements for publication. Repeated narrow supervision may limit generalization, but the study does not isolate overfitting as the cause of its errors.

VII. RQ3: SYSTEMS AND PRIVACY OVERHEAD

A. Measured Systems Cost

Historical V1 timings in Table XI end at the local rules/NER decision. All 240 ALLOW/WARN decisions ran both detectors, whereas the 120 BLOCK decisions stopped after rules, legitimately skipping NER. The cells therefore include different amounts of work. Medians span 27.9–35.0 ms, and each maximum describes sixty requests across two profiles, excluding receiver completion and semantic processing.

Across 126.40 seconds of resource windows, the container accumulated 193.29 CPU-seconds, including 0.69 observer CPU-seconds; CPU time sums concurrent processor use. Maximum charged memory reached 1,267.91 MiB for services, cache, driver and receiver together. Sampling gaps of up to 622.84 ms could miss shorter spikes. Final-read and cleanup checks passed, but the combined resource totals cannot isolate incremental overhead.

Installed V3 balanced timings appear in Table XII. The four within-profile median increments per arm span 26.1–26.7 ms for rules/NER, 9.4–17.0 ms for semantics and 31.9–33.5 ms for full detection; these compare profile medians rather than individual requests. All 480 timed attempts returned ALLOW. Including 80 warmups, 144 scaffold calls and ten post-cost checks, the 714 calls yielded 574 analysis RPCs and 712 unchanged-body arrivals without execution errors. Quality blocked the same benign prompt once on each transport, accounting for both non-arrivals. Correct package loading thus coexisted with an inappropriate decision, although one repeated prompt cannot establish a false-positive rate.

TABLE XI
PAGE-OBSERVED RULES/NER DECISION LATENCY (MS), WITH SIXTY MEASUREMENTS ACROSS TWO FRESH PROFILES PER CELL. MAX IS THE OBSERVED MAXIMUM; TIMINGS EXCLUDE RECEIVER RESPONSE AND SEMANTIC PROCESSING.

Transport	Action	p50	p95	Max
fetch	ALLOW	35.0	62.5	73.6
fetch	WARN	31.9	36.1	59.4
fetch	BLOCK	27.9	31.6	33.2
XHR	ALLOW	30.8	35.8	57.7
XHR	WARN	31.3	34.6	59.2
XHR	BLOCK	28.5	32.2	34.1

TABLE XII
INSTALLED V3 BALANCED REQUEST-TO-RECEIVER LATENCY (MS), WITH SIXTY TIMED REQUESTS ACROSS TWO PROFILES PER CELL. OFF RETAINS EXTENSION/NATIVE INFRASTRUCTURE, AND OBSERVED MAXIMA DO NOT GUARANTEE TAIL PERFORMANCE.

Transport	Detectors	p50	p95	Max
fetch	Off	3.2	4.0	4.3
fetch	Rules/NER	29.6	33.8	92.4
fetch	Semantic	19.7	158.8	283.6
fetch	Full	35.4	172.8	202.1
XHR	Off	3.3	5.0	7.4
XHR	Rules/NER	29.9	31.8	100.5
XHR	Semantic	15.1	162.8	312.7
XHR	Full	35.9	175.8	341.0

These comparisons isolate detector activation for one short input, excluding total overhead relative to an absent extension and the cost of cold download or loading. With only two profile clusters per cell and observed maxima up to 341.0 ms, they establish neither a service-level objective nor a deployment-tail guarantee. Product code, dependencies and images also differ between V1 and V3, preventing attribution of cross-study speed differences to the model alone. Both studies timed the pre-change rules and NER; case-preserved input, the added validators and the NER post-filter were not part of the timed arms, so the rules/NER increments may differ for the installed configuration. A single sequential detector-layer pass (Section V-C) found no meaningful per-prompt latency difference between configurations, but it is not a browser measurement.

B. Privacy Cost and Telemetry Utility

The finite audit qualifies the privacy guarantee suggested by the ideal formula. On the fingerprinted host, maximum

TABLE XIII

TWO ILLUSTRATIVE PRESPECIFIED CATEGORY CELLS ($K = 11$, EVENT $\varepsilon = 1$), EACH REPEATED 10,000 TIMES. TRUTH, CLIPPED BIAS AND RMSE DESCRIBE THE FIRST CATEGORY IN COUNT UNITS. CLIP GIVES ITS CLIPPING FREQUENCY; SUM GIVES THE FREQUENCY OF INDEPENDENTLY CLIPPED MARGINALS FAILING TO SUM TO n (TOLERANCE 10^{-9}).

n	Truth	Bias	RMSE	Clip (%)	Sum (%)
80	7	+22.84	42.32	85.62	99.99
8000	727	+237.50	1028.47	28.32	99.78

event loss exceeded nominal budgets 0.5, 1 and 2 by approximately $(1.294, 1.503, 1.226) \times 10^{-15}$. Although tiny, this excess means the kernel does not satisfy the exact nominal bound. Eight default reservations compose to approximately $8 + 1.202 \times 10^{-14}$ under the same independent uniform-entropy assumptions. The result bounds admitted event content under the audited conditions, without certifying other builds or deployment entropy.

The two prespecified balanced-integer category cells in Table XIII, at event budget one and $n = 80/8000$, illustrate clipping bias and root-mean-square error (RMSE). They do not establish utility acceptance or describe every cell. Absolute error grows with contributor count even when relative error can decline, and independently estimated marginals almost never sum to the total. Additional reports alone therefore do not ensure reliable counts; acceptable application error and deployment accuracy remain unmeasured.

Accounting checks admitted eight default reservations, denied the ninth, preserved the cap after reopening and reset it on the next UTC day. Clock rollback and corrupt state failed closed; sixteen of forty concurrent epsilon-0.5 requests were admitted. Construction, queue and delivery failures still consumed reservations. These checks support bounded admission under the tested state and clock conditions, but admission also determines which events are represented. In a constructed 32-event sequence containing 25% BLOCK overall, reversing the order changed BLOCK representation among the first eight contributors from 100% to 0%. Content randomization cannot correct that selection or conceal participation, so a content-privacy bound alone cannot establish representative monitoring.

The controlled operational run offered 308 opportunities across 23 ledgers, producing 184 reservations, 168 accepted reports, sixteen submission failures and 124 pre-queue suppressions. Reopen, recovery and invalid-packet checks passed, and noisy and estimated summaries survived collector restarts. Ordered ALLOW-then-BLOCK inputs exhausted budgets before BLOCK, reproducing the selection effect in fixtures rather than users. Across all 308 opportunities, including suppressions, synchronous report-call p50/p95/max was 56.157/143.356/189.324 ms. Reporting occurs before response return but after construction of the runtime elapsed field, so that field excludes reporting cost. Separate operator Health probes detected outage and recovery without demonstrating autonomous monitoring.

Answer to RQ3. Installed measurements quantify resident V3 detector activation alongside historical infrastructure costs, while leaving total and cold-start overhead unresolved. Controlled telemetry tests verify delivery and accounting, but finite privacy excess, estimation error and admission selection persist. The channel lacks performance fields, and useful deployment monitoring remains unmeasured; its content guarantee also excludes participation and network metadata.

VIII. DISCUSSION AND LIMITATIONS

A. What the System Evidence Supports

Detector placement is part of the policy decision, not simply an implementation choice. An early rule-based BLOCK prevents semantic contribution, so a better downstream model cannot repair it. Historically, removing rules corrected 139 unnecessary restrictions but lost eighteen previously correct actions. The appropriate response is therefore to examine both restriction and protection costs, rather than infer that higher aggregate agreement justifies removal. This implication remains conditional on the provisional policy and tested fixture, and both counts come from the pre-change rules. The precision repairs of Section III-C were designed to reduce such restrictions, and the full-product rerun of Section V-D found that they did so in part: the shared unnecessary restrictions fell from 139 to 108 without new binary misses. Removing rules nevertheless still corrected more actions than it lost (net 85) at the cost of the same eighteen lost actions, so the tradeoff survived the repair. The registered fresh evidence adds a second tradeoff at the detector layers (Section V-E). The revisions raised union specificity by about three points and reduced over-masking, and they cost about one point of union recall, concentrated in one source and in the NER layer, while the rework restored BLOCK-level recall. The recall cost is real, and its size relative to the specificity gain does not settle whether the trade is acceptable. That is a product decision, which the owner took by shipping the rework; the registered rule did not support promotion, and the specificity gains rest partly on second-look negatives. A policy-correct WARN presents a related distinction: because it releases the original content, classification correctness, appropriate intervention and prevented transmission each require their own evidence.

Comparison with local sanitization clarifies who controls continuation. Casper's acknowledged warnings leave a decision to the user, while HaS and ProSan transform content to support continued use [11], [13], [14]. PriVoke instead proceeds after WARN without acknowledgement and leaves task completion unresolved after BLOCK; it neither masks prompts nor restores responses. Receiver compliance verifies that contract, but not its desirability. Establishing the latter would require evidence about authorized task completion, warning comprehension and the effort needed to reformulate a prompt.

The ablation and execution traces provide complementary explanations. Traces identify where a decision arose and show when downstream layers did not execute. The matched removal changes one detector configuration and observes the

resulting paired actions on the same fixture. The trace alone cannot establish what the removed layer would have changed; the removal result alone does not establish that a new policy would be acceptable to users. Together they localize a tradeoff that a future policy study would need to resolve. In particular, fewer unnecessary restrictions and more missed protections should remain visible as different consequences, rather than being collapsed into a recommendation to remove the rules.

B. Ground Truth and Generalization Remain Open

Contextual integrity concerns appropriate information flows [8], for which entity recognition supplies only part of the judgment. Independent contextual ground truth remains the largest scientific gap because PriVoke's authored targets are assistant-provisional. Freezing the separate annotation-presence endpoint before opening limits post-selection, but neither its different target nor its sole-semantic execution validates contextual authorization. Improved authored-target agreement and poor final specificity consequently answer different questions, neither establishing human agreement, appropriate intervention or task utility.

A stronger contextual evaluation would specify who may disclose which information to whom, for what purpose, before annotators see model outputs. It would need to preserve legitimate disagreements rather than force every situation into a single unexamined target. The current allowed-action sets accommodate some policy latitude, but their provisional construction is not a substitute for that review. The annotation-presence endpoint is still informative: it exposes behavior on a separately frozen target. Its limitations do not justify dismissing adverse specificity, just as its frozen status does not convert it into evidence of contextual authorization.

Group and text exclusions reduce declared overlap, while paired cluster analyses show sensitivity to known dependence. Neither establishes template independence, paraphrase exclusion or absence of pretraining exposure. The final endpoint comprises 308 Nemotron, 89 AI4privacy, eighty Gretel, twenty Privy and one MAPA row. These English synthetic or machine-generated cases do not represent other languages or organizational policies, and neither unadjusted intervals nor three seeds provide a population guarantee. Confirmation after opening this endpoint requires a new protected successor.

The detector-precision unit adds its own limits. It uses a single small English spaCy model, and its cue lists for honorifics, relations and roles are English-only. Annotation-presence labels are source annotations with unquantified noise, and a prediction is positive for any nonzero sensitivity, so S1 findings count as positives. The filter was designed on dev-TUNE and the personalized set; dev-CHECK and the OpenAI set were scored before the post-review revision and are no longer strictly held out, so only the one-shot run on the final endpoint is free of selection on its rows, and that endpoint was already opened for unit 3 and has since been used a second time, for paired statistics on the same frozen pair, so it cannot inform further decisions. The student-identifier rule was added after the endpoint scoring and rests on a single template, and only the fresh held-out evaluation of Section V-E includes it. The datasets are synthetic or benchmark-derived, and small splits make single rows move recall by about a percentage point. Pets, fictional or historical people and organizations that resemble given names cannot be separated from private people without world knowledge. The held-out run has 236 positive and 262 negative rows, so a single row moves recall by about 0.4 points and specificity by about 0.4 points; its regex differences are net differences of three rows and are not statistically significant. Percentile bootstrap intervals are optimistic when few rows change, so the exact tests are the primary evidence for small counts; no multiplicity correction was applied, and the synthetic sets lack template identifiers, which makes their row-level inference likely anti-conservative. Timing is one sequential pass on one host.

The fresh held-out evidence has its own limits. F4 contains positive sentences only, so K's specificity rests on negatives that O and N had already been scored on and whose results informed the rework, and on unlabelled natural prompts whose flag rates cannot be read as error rates. The benchmark pool is exhausted, so no further confirmation draw is possible without a new source. Its annotation is incomplete by construction and includes attribute spans (age, title, gender) for which PriVoke has no detector, which is why recall is far from 100%, and label noise was not audited. The F2 adjudication and the label-noise audit remain unperformed. The K qualifier vocabulary mirrors labels present in the development spans, so labelled-identifier gains are tied to the benchmark's labelling conventions. The registered promotion rule was not met, and K is shipped by owner decision. K's recall cost against the original detectors is real, source dependent and partly attributable to the NER layer it shares with the first revision. Exploratory layer, rule, stratum and span analyses were produced after the registered results and cannot confirm anything. Neither the full-product rerun nor the installed probes and cost studies include K, and the default-off robustness fixes and larger NER models are not part of the shipped configuration. All detector-layer results use one small English spaCy model and annotation-presence labels, not contextual authorization.

C. Retention, Availability and Practical Utility

Matching the base and exposure reduces RQ2 configuration confounding without retrospectively isolating the RQ1 package effect. Consistent with negative-flip research [10], all matched full-model candidates improved aggregate assessment yet failed the guard. Under the configured rule, higher scores cannot override casewise rejection, and no candidate was published. The weak zero-head base and small guard also constrain the significance of a pass, motivating independently reviewed retention cases and explicit criteria for harm.

The meaning of a harm criterion matters as much as its pass rate. A label change that preserves an allowed action differs from a change that releases previously blocked content; an unnecessary BLOCK imposes a different cost from under-restriction. The maintained guard uses a single derived action and the assessment uses allowed sets, so their disagreements

need not be contradictions. A larger, independently justified guard could alter eligibility in either direction. The present result establishes the configured decision and the measured gains and losses, without establishing that the guard represents every relevant harm or that rejected weights lack all useful information.

Operational version checks address another failure mode. Binding a candidate to its exact parent can prevent an update evaluated against one model from overwriting a different revision, but cannot make its labels correct. Likewise, a verified download shows artifact integrity, not behavioral suitability. Standard V3's pinned inference path and the separately enabled legacy update service must remain distinguishable: this paper reports installed V3 inference and bounded continuation evidence, not a deployed V3 learning loop.

Blocking when analysis is unavailable can prevent disclosure while interrupting an authorized task; WARN avoids that interruption but proceeds without informed acknowledgement. Heuristic confidence cannot resolve the choice because it is not calibrated joint correctness. Installed Linux evidence now covers historical V1 enforcement and standard V3 loading and cost, yet the quality-tier benign-prompt BLOCKs demonstrate why installation alone is not accuracy validation. Authenticated providers, Windows, workers, attachments and other transports remain unverified. Without evidence on authorized task completion, unnecessary intervention, warning comprehension and reformulation effort, practical utility and human benefit remain open.

D. Measurement Boundaries and Next Steps

Matched resident arms quantify semantic and full-detector activation, but retain instrumentation and omit extension-absent and cold-load comparisons needed to establish total overhead. For reporting, RAPPOR motivates repeated-report accounting [16]; PriVoke's ledger depends on retained state and a trusted clock, and its finite bound assumes independent uniform OS bits. Accounting leaves estimation error and admission-order effects unresolved, so privacy, estimation and representative admission require separate justification. The current deployment has too few concurrent users for a live-usage usefulness study, and more users alone would not repair the missing fields. Lifetime categorical marginals and an operator Health probe cannot replace latency/error and usage-denominator instrumentation, nor has a deployment-usefulness threshold been established. Participation, delivery selection and metadata remain outside the bound, and aggregates supply no corrected training labels.

A reporting design must consequently specify its intended estimand before increasing collection. The distribution among admitted, successfully delivered events can differ from the distribution among all attempted decisions, installations or users. A daily cap selects early events; delivery failures can further select which reservations become visible. Even an exact estimator of the admitted histogram would not correct those differences without additional assumptions. Adding richer fields or denominators could help answer operational questions, but would change both the disclosure surface and the privacy analysis. Such a redesign and its validation are future work, rather than benefits established by the current collector.

Reproducibility has a related boundary. A readable recipe and a fingerprint allow an available artifact to be identified, while a successful fresh rerun additionally requires the referenced weights, inputs, compatible dependencies and permissions. The retained records disclose local and ignored assets so that absence from a public checkout is not mistaken for a complete reproduction package. Future release work should resolve distribution and environment requirements independently of the manuscript's empirical conclusions.

A human double-annotation protocol for contextual targets, with blinded stratified sampling and agreement statistics, has been prepared but not run; running it, together with adjudicating a flagged sample of natural prompts, is the most direct route to independent evidence for the detector findings above.

Across the lifecycle, early restrictions can survive later model improvements, improved candidates can remain ineligible, and admitted reports can remain unrepresentative. These findings motivate separate acceptance evidence rather than validate a remedy. Contextual adjudication, broader receiver and retention checks, total-cost measurements, and user/task evidence are needed to support stronger claims.

IX. Conclusion

PriVoke integrates local prompt inspection, explicit request enforcement, version-bound guarded revision and independently randomized reporting. Its evaluation demonstrates why each mechanism needs distinct acceptance evidence. Agreement with authored contextual targets improved, while action errors and poor specificity on a separate annotation-presence task persisted. Installed probes verified supported request handling, including unchanged-content forwarding under WARN. All matched full-model candidates improved aggregate agreement but failed retention, making average gain insufficient to justify publication. Installed V3 tests measured resident activation cost while revealing an adverse quality-tier decision; controlled telemetry verified delivery and accounting without establishing live monitoring usefulness. The contribution is an inspectable lifecycle and an empirical account of its limits. A detector-layer precision revision shifted held-out regex specificity upward and recall downward by amounts too small to distinguish from chance (three rows each), with significant changes only on synthetic sets and no measured net NER benefit. A full-product rerun found fewer unnecessary restrictions (139 to 108) without new binary misses, yet the rule-removal tradeoff persisted and the NER layer began converting findings. Registered tests on fresh sentences then showed that the first revision cost 1.37 points of union recall for 3.3 points of union specificity, and that the shipped rework, which restored BLOCK-level recall, still cost 1.05 points (interval -1.70 to -0.41) against the original detectors, concentrated in one source and in the NER layer. It was shipped by owner decision, not by meeting the registered promotion rule.

Independent contextual accuracy, useful adaptation without lost protection and human benefit remain to be established.

AI Assistance

Artificial intelligence (AI) assistance supported provisional contextual labelling and manuscript drafting. GPT-6 Sol assisted implementation and controlled testing, GPT-6 Luna assisted review, and GPT-6 Astra assisted manuscript review and rewriting. These provisional labels have not been independently adjudicated as ground truth. The authors remain responsible for the manuscript and its claims.

Appendix A
Policy Decision Order and Observable Outcomes

The contextual attributes in Section III-B describe a finding before enforcement. For a finding with numeric confidence below 0.5, the low-confidence rule takes precedence: S3 yields WARN; S2 yields WARN only if IDENTITY or LOCATION is present; all remaining cases yield ALLOW. Missing or nonnumeric confidence does not activate that rule. This is a decision threshold, not an empirical confidence calibration.

For other findings, the baseline checks are ordered. S0 with an empty category set yields ALLOW. S3 yields BLOCK and S2 yields WARN. For the remaining S0/S1 cases, IDENTITY together with LOCATION yields WARN; either category with visibility P2, P3 or P4 also yields WARN. Otherwise the finding yields ALLOW. PU denotes an unspecified audience and is not treated as restricted/private by this comparison. A category set is multilabel, so retaining one sensitive category is not equivalent to matching the exact target set.

The cascade then constrains what reaches aggregation. An early rule BLOCK ends inspection; later requested layers have not evaluated the text. Otherwise, aggregation preserves the strongest enforcement decision rather than replacing it with the last detector's action. A failed requested layer changes an otherwise ALLOW result to BLOCK, while a valid non-ALLOW finding remains visible. These semantics differ from a configuration in which an unavailable parameter stream leaves fewer enabled layers: a successfully completed reduced cascade is not a successful semantic execution.

Finally, a policy action and a receiver observation are different records. ALLOW and WARN release the original supported request, while BLOCK cancels it. Notice rendering under WARN does not imply acknowledgement, consent or removal of information from the body. For a probe expecting a send, the relevant observation is receiver arrival with the original body; for a no-send probe, the observation is absence of arrival within the declared observation window, including checks for a late send. The installed probes establish these bounded outcomes, not an absence of leakage through every browser transport.

Appendix B
Measurement Definitions and Reused Evidence

Let case j have target sensitivity s_j, audience v_j, category set C_j and allowed-action set $\mathcal{A}_j$. Let the corresponding prediction be $\hat{s}_j, \hat{v}_j, \hat{C}_j, \hat{a}_j$. Define the three indicators

$$L_j = \mathbf{1}\{\hat{s}_j = s_j,\ \hat{v}_j = v_j,\ \hat{C}_j = C_j\}, \tag{3}$$

$$A_j = \mathbf{1}\{\hat{a}_j \in \mathcal{A}_j\}, \tag{4}$$

$$J_j = L_j A_j. \tag{5}$$

Label agreement, action agreement and label-and-action joint agreement average L_j, A_j and J_j, respectively, over the declared cases. Exact category agreement counts both omitted and spurious categories as mismatches. For action direction, the ordering is ALLOW $<$ WARN $<$ BLOCK; the retained contextual analysis distinguishes an action below the allowed set from one above it.

The binary projection is positive when $\hat{s}_j \neq \mathrm{S0}$ or $\hat{C}_j$ is nonempty. Recall is $\mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$ and specificity is $\mathrm{TN}/(\mathrm{TN} + \mathrm{FP})$, with denominators fixed by the declared positive and negative targets. On the contextual workload those targets are provisional sensitivity labels; on the final endpoint they are source annotation presence. The same prediction projection does not make those reference tasks equivalent.

For any binary correctness indicator Z_j, the paired gain count is the number of cases with $Z_j^{\mathrm{old}} = 0$ and $Z_j^{\mathrm{new}} = 1$; the loss count reverses those values. The net rate difference is $(G - L)/N$. Reporting only that difference would hide the number of previous successes lost. The case identity must remain paired when a cluster is resampled, and repeated appearances of a sampled group carry its rows with their multiplicities. Row weighting is retained, so a larger group contributes more cases; the method does not give each group equal influence.

The 1,560 contextual cases are shared by the package comparison and targeted ablation. The 1,080-case continuation assessment is a reused provisional subset after the declared exclusions. These are related development observations. The 498-row final endpoint supplies a different frozen target and source composition, with all six arms retained, but its 464 observed group keys do not certify independent templates. A later analysis of already opened material cannot retroactively make it protected confirmation. The full confusion counts and adverse outcomes remain in the main results, rather than being displaced by this methodological detail.

Appendix C
Candidate Identity, Guarding and Reproduction

The matched continuation comparison starts both HEADS and FULL from the same intact pretrained balanced encoder/pooler and zero task heads. That ORIGINAL control is distinct from both the historical V1 classifier and the selected updated package used in RQ1. The shared 4,096 exposures describe repeated training occurrences, not 4,096 independently sampled situations. Each seed produces a terminal candidate; the assessment selects no intermediate checkpoint.

A candidate, a guard decision and an effective artifact must remain distinguishable in the record. The candidate is the fitted set of weights, even if rejected. The decision applies the fixed sixteen-case criterion: improved joint correctness and zero action harms against its derived target actions. The effective-artifact mapping describes which artifact that rule would retain: a rejected candidate maps to the unchanged ORIGINAL. Because this experiment publishes no candidate, that mapping does not establish a server replacement or a client refresh. Counting only effective artifacts would erase the rejected candidates' measured gains and harms; counting every fitted candidate as an update would erase the guard.

For reproduction, the maintained records identify the exact base and candidate artifacts, parameter scope, ordered exposure schedule, exclusions, scoring target and actual sole-semantic execution. The model-family label alone is insufficient to recover those inputs. The same requirement applies to the RQ1 package comparison and final endpoint: source, tokenizer, backbone and task-head identities must match the recorded configuration. Some checkpoints and raw run receipts remain local and ignored, so the public repository does not by itself establish that every input can be obtained and redistributed.

The repository locators for these retained units are `paper/research/controlled-studies-20261014.md`, `pipeline-comparison-20261014.json` and `final-semantic-evidence-20261014.json`. The dated records preserve their historical source closures and failed preparation attempts. They provide provenance for the reported comparisons; recovering their inputs does not change the original selection and assessment conditions.

Appendix D
Installed-Browser Observation Boundaries

The historical rules/NER cost capture and standard-V3 activation study use different timing endpoints. Historical page-observed decision latency ends when the analysis result returns and excludes the receiver response. V3 request-to-receiver completion includes a successful send to the synthetic receiver. Their distributions should therefore not be subtracted to estimate a generation speedup. In the V3 study, subtraction occurs only within a profile between its resident detector arm and its detectors-off baseline, using the same endpoint and fixed prompt.

The off baseline retains the extension hooks, native launcher, supervisor, bridge and model service. It estimates the increment from activating a detector in that arrangement, not the total cost of installing the extension. Reversing the arm order in the second profile replicate reduces reliance on one fixed ordering without creating an extension-absent control or a cold-load experiment. The warmup requests are excluded from the timed distributions, and the untimed tier-selection probes remain separate from those distributions.

Historical resource windows include browser startup and request/readiness activity before teardown, with resident services and instrumentation already present. Cgroup CPU counters aggregate descendants, while charged memory includes cache and kernel charges [22]. They are not per-detector CPU or process RSS/PSS measurements. Positive sampling gaps remain part of the record; a sampled maximum does not establish the continuous peak. Observer cost is retained rather than subtracted through an unverified attribution model.

Each historical transport/action cell pools thirty measurements from each of two profiles. Its nearest-rank p99 is the observed maximum of sixty measurements, rather than evidence of a well-estimated population tail. The V3 distributions likewise contain repeated executions within only two profiles per transport. These restrictions matter when interpreting apparent latency differences and when extrapolating beyond the fixed synthetic prompts. The installed evidence records are `installed-client-evidence-20261014.md` and `v3-standard-installed-evidence-20261017.md` under `paper/research/`.

Appendix E
Telemetry Estimation and Admission

The ideal mechanism and audited finite implementation in Section III-E are analyzed for the content of a fixed reporting opportunity under public configuration. The finite bound includes the reported excess over nominal privacy loss. For a category whose true contributor count is h among n reports, and observed randomized count Y, the nominal inverse-response estimate before clipping is

$$\widehat{h} = \frac{Y - nq}{p - q}. \tag{6}$$

The collector clips each estimated marginal separately to $[0, n]$. Inverting the nominal probabilities is not the same as inverting the exact finite kernel, and clipping is nonlinear: the expected clipped value need not equal h. Independently clipped categories also need not sum to n. These properties explain why privacy loss, bias, RMSE and total coherence are separate checks.

The synthetic trials fix the contributor histogram before sampling the randomized reports. Their error therefore excludes uncertainty about which people or events enter that histogram. Increasing n means more admitted reports, pooled across installation-days where necessary, rather than relaxing the per-installation daily cap. A precision requirement for an operational decision was not prespecified; the two illustrative category cells cannot establish one after the fact.

The operational accounting also has distinct denominators. An opportunity may be suppressed before queuing, a reservation may be consumed without a delivered packet, and an accepted report represents a received randomized event. The observed totals reconcile as $308 = 184 + 124$ opportunities and $184 = 168 + 16$ reserved events, where the latter partition is accepted reports plus submission failures in the controlled fixtures. These are not counts of users, unique prompts or detector decisions in a deployment.

The ledger reserves before construction and does not refund failed construction or delivery. That ordering protects accounting against retries within retained state, but cannot hide whether a report was sent. A cap can also change event composition before randomization, as the retained order-reversal example demonstrates. Correcting the randomized counts would still leave that admission mechanism to justify. The finite audit and controlled operation are recorded separately in `telemetry-evidence-20261014.md` and `rq3-telemetry-evidence-20261017.md`; neither supplies training examples or establishes useful live monitoring.

Appendix F
Detector Precision Rules and Reproduction

This appendix defines the controls evaluated in Sections V-C and V-E. The records `detector-precision-20261024.md` and `detector-precision-heldout-20261025.md` under `paper/research/` hold source revisions, split definitions and results.

Rule gates. Kinship terms and the word "family" require a possessive determiner (my, our, his, her, their), optionally followed by up to two hyphen- or space-joined modifier words, so "my ex-wife" matches while a non-possessive self-statement does not. Child cues require a possessive and a child noun, an age expression, or an age followed by a person noun; object ages such as a five-year-old laptop do not match. Financial keywords cover personal financial-state nouns (for example salary, loan, mortgage, savings, credit score) after a possessive with up to two modifiers; a bare reference to an account does not match. Location keywords require a connector such as "live in" followed by a capture that is not a closed-class word. Structured identity values stay on one line and capture at most five tokens, and a phone label needs a value that begins like a number. Checksum validators reject shape-only hits for card numbers (Luhn), IBANs (mod-97), NHS numbers (mod-11) and US social security numbers (issuable area, group and serial ranges), and context matching is word-bounded rather than substring-based. Password placeholders, chat mentions and standards identifiers are excluded. A bare reference to a student is not a child cue; a separate student-identifier rule (S3 identity) needs a value of four or more characters containing a digit tied to the word student, so generic student and identifier mentions stay negative. It postdates the measurements of Section V-C and Section V-D and is part of N and K in Section V-E.

Labelled identifiers and downgrades (K). Two rules, S3 and S2 identity, match a label followed by a value. The label noun is id, identifier, number, no, num, nr, code or reference, in words, snake case, camel case or key-value form, or a hash sign directly after the qualifier. S3 qualifiers are passport, licence, national, social security or insurance, tax, identity, identification, citizenship, voter, visa, residence permit, immigration, military, birth certificate, patient, medical record, health plan, insurance, Medicare, Medicaid, beneficiary, bank account and payment card; the acronyms SSN, TFN, NINO, EIN, ITIN, MRN and NRIC need no noun, and a colon after Passport, ID card, identity card, Medicare or Medicaid labels its own value. S2 qualifiers are customer, client, account, member, subscriber, employee, staff, personnel, user, policy, claim, registration, plate, store, loyalty, library, rewards, club and gift cards, and reference. A value is tokens of letters, digits and underscores joined by hyphens or single spaces; its first token carries a digit or is a one- to three-letter prefix before a digit group, every spaced group carries a digit, it contains five to twenty-four compact characters and it is not a date. Software licence names, order, build, ticket, part and promotion codes are not qualifiers. Downgrades keep a hit whose checksum fails when a card, banking or social-security word is in context (S3) and, after an IBAN check fails without a banking word, keep a token of eight or more characters with five or more digits and a letter at S2. Because digit spaces are removed before matching, a spaced card number reaches the rule as one digit run.

Post-filter, stage 1. Five span rules apply in turn. The placeholder rule removes masks (repeated letters, spans without letters, a pluralized capital letter, and spans inside template delimiters). The identifier rule removes e-mail, URL, underscore and code-like tokens, which belong to the regex layer. The structural rule removes spans inside headings, except multi-token or cue-introduced person names. The short-acronym rule requires a locating or affiliation preposition before a place of at most three letters or an organization of at most four. The common-phrase rule removes a single-token person candidate when the model's installed word list contains it and the tagger, reading the lowercased token in isolation, finds no proper-noun reading and no context cue corroborates it.

Post-filter, stage 2. The first matching rule decides: (1) a person linked with a place or organization, when either side is independently corroborated, yields S2; (2) the corresponding place or organization yields S2; (3) a corroborated person yields S2; (4) a place or organization tied to the speaker by a relation cue keeps its default sensitivity; (5) one in the same prompt as such a tied entity keeps its default; (6) a facility or location directly beside a number yields S2; (7) a single uncorroborated person yields S1; and (8) a lone place or organization is dropped. Independent corroboration means evidence about the entity itself: a multi-token, cue-introduced or recurring person, or a place or organization with a personal-relation cue. A bare locating preposition does not corroborate.

Development-only ablations. Table XIV holds the repaired rules fixed and varies only the NER input and filter, using union recall/specificity from the source repository's experiment records. These arms were run on design and development splits; dev-CHECK and OpenAI are post-review, and none is held out. Case preservation alone raises or preserves union recall but lowers specificity on every split with a margin to lose (to 0.824, 0.832 and 0.814 on dev-TUNE, dev-CHECK and OpenAI, and to 0.919 on the personalized set), because capitalized non-names enter the NER output. The post-filter recovers most of that specificity on case-preserved input, and on lowercased input it lowers recall slightly while raising specificity. Counting only non-organization entities as NER

TABLE XIV
DEVELOPMENT-ONLY UNION ABLATION (RECALL/SPECIFICITY), REPAIRED RULES FIXED. CASE = CASE-PRESERVED NER INPUT; PF = POST-FILTER (REVISED VERSION). †POST-REVIEW SET. NOT HELD OUT.

NER arm	Tune	Check†	Personal.	OpenAI†
Lowercase	.757/.966	.681/.950	1.000/1.000	.926/.916
Case	.858/.824	.819/.832	1.000/.919	.942/.814
Case + PF	.797/.950	.716/.950	1.000/1.000	.918/.926
Lowercase + PF	.736/.983	.664/.975	1.000/1.000	.914/.926

positives moved development NER-only recall from 0.170 to 0.333 and specificity from 0.996 to 0.937 between lowercased and case-preserved input.

Reproduction. The evaluation script `evaluation/evaluate_installed_detectors.py` calls the analysis pipeline per prompt with the explicit layer list regex and NER (never empty), on the host Python environment with spaCy `en_core_web_sm`, and records the runtime flag values it observed. The pre-change configuration is a clean export of commit `e6c177ff`; the installed configuration is source commit `89c358fd`, which differs from the defaults-on commit `6a0c9ca3` only by that script. The final endpoint was scored once with an explicit one-shot option. Dataset SHA-256 digests, the full commit list and the aggregates are in `detector-precision-20261024.md`.

Follow-up provenance. The paired statistics come from branch `research/wt-a-stats`, commit `bf5e24c1`, which adds per-row detector outputs and a seeded grouped bootstrap; the pre-change tree is the clean export of `e6c177ff` and the revised tree is `55cbd1e5`. The full-product rerun comes from branch `research/wt-h-product`, commit `91b4ee83`, on the same two trees, with the frozen historical protocol, package checksums and 1,560-row input digest re-verified before each run. The student-identifier rule is commit `e70e3a2e` on `fix/regex-fp-reduction` and is documented in its own experiment record (`42f8f529`). The shipped rework K is code freeze `b7e716c3`, with the development-data record at `c8ceaacc`, the head of pull request 5 on the same branch. The record `detector-precision-heldout-20261025.md` lists the branches and commits of the fresh held-out registrations and the supplementary checks, and the exact prompts.

APPENDIX G
FRESH HELD-OUT DETAILS

Table XV gives union-recall deltas in the registered source strata. Sources are the benchmark's component datasets. Strata are secondary analyses and were not used for promotion. Row counts are positive sentences in F1 and F4.

On the F4 BLOCK-level cell, K−O is negative in ai4privacy (−1.05 [−1.60, −0.52]) and Privy (−1.12 [−1.58, −0.70]) and positive in Gretel (+1.47 [+0.81, +2.13]) and Nemotron-PII (+2.05 [+0.44, +3.74]), whereas the pooled estimate is level. In the F1 strata, the larger spaCy model raised union specificity over N in ai4privacy, Gretel and Nemotron-PII and lowered it in Privy. The five G2 losses all came from the generic labelled-identifier rule. Runs were scored in the host Python environment with spaCy 3.7.5 and `en_core_web_sm` 3.7.1; the freeze records fix hashes of the sets, trees and scoring scripts, and no hypothesis, margin or code path changed after the freeze.

REFERENCES

[1] B. B. Gupta, A. Gaurav, V. Arya, W. Alhalabi, D. Alsalman, and P. Vijayakumar, "Enhancing user prompt confidentiality in large language models through advanced differential encryption," *Computers & Electrical Engineering*, vol. 116, p. 109215, 05 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0045790624001435

[2] W. Liang, Y. Zhang, M. Codreanu, J. Wang, H. Cao, and J. Zou, "The widespread adoption of large language model-assisted writing across society," *Patterns*, vol. 6, no. 12, p. 101366, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2666389925002144

[3] T. Koraag, N. Wagner, F. Dobslaw, and L. Gren, "Comparing open-source and commercial LLMs for domain-specific analysis and reporting: Software engineering challenges and design trade-offs," *arXiv preprint arXiv:2509.24344*, 2025. [Online]. Available: https://arxiv.org/abs/2509.24344

[4] S. Ray, "Samsung bans ChatGPT among employees after sensitive code leak," *Forbes Business*, 05 2023. [Online]. Available: https://www.forbes.com/sites/siladityaray/2023/05/02/samsung-bans-chatgpt-and-other-chatbots-for-employees-after-sensitive-code-leak/

[5] J. Dastin and A. Tong, "Focus: Google, one of AI's biggest backers, warns own staff about chatbots," *Reuters (Technology)*, 06 2023. [Online]. Available: https://www.reuters.com/technology/google-one-ais-biggest-backers-warns-own-staff-about-chatbots-2023-06-15/

[6] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, A. Rodriguez, A. Joulin, E. Grave, and G. Lample, "LLaMA: Open and efficient foundation language models," 02 2023. [Online]. Available: https://arxiv.org/abs/2302.13971

[7] A. Priyanshu, S. Vijay, A. Kumar, R. Naidu, and F. Mireshghallah, "Are chatbots ready for privacy-sensitive applications? an investigation into input regurgitation and prompt-induced sanitization," *arXiv preprint arXiv:2305.15008*, 2023. [Online]. Available: https://arxiv.org/abs/2305.15008

[8] H. Nissenbaum, "Privacy as contextual integrity," *Washington Law Review*, vol. 79, no. 1, pp. 119–158, 2004. [Online]. Available: https://digitalcommons.law.uw.edu/wlr/vol79/iss1/10/

[9] I. Turc, M.-W. Chang, K. Lee, and K. Toutanova, "Well-read students learn better: On the importance of pre-training compact models," 2019. [Online]. Available: https://arxiv.org/abs/1908.08962

[10] S. Yan, Y. Xiong, K. Kundu, S. Yang, S. Deng, M. Wang, W. Xia, and S. Soatto, "Positive-congruent training: Towards regression-free model updates," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021, pp. 14 299–14 308. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2021/html/Yan_Positive-Congruent_Training_Towards_Regression-Free_Model_Updates_CVPR_2021_paper.html

[11] C. J. Chong, C. Hou, Z. Yao, and S. M. S. Talebi, "Casper: Prompt sanitization for protecting user privacy in web-based large language models," 08 2024, version 1, 13 August 2024; supplied 13-page PDF. Not the 2025 proceedings version. [Online]. Available: https://arxiv.org/abs/2408.07004v1

[12] C. J. Chong, C. Hou, Z. Yao, and S. M. S. Talebi, "Casper: Prompt sanitization for protecting user privacy in web-based large language models," in *2025 IEEE 12th International Conference on Cyber Security and Cloud Computing (CSCloud)*, 2025, pp. 122–131, related preprint: arXiv:2408.07004v1. [Online]. Available: https://doi.org/10.1109/CSCloud66326.2025.00027

[13] Y. Chen, T. Li, H. Liu, and Y. Yu, "Hide and seek (HaS): A lightweight framework for prompt privacy protection," 09 2023. [Online]. Available: https://arxiv.org/abs/2309.03057

[14] Z. Shen, Z. Xi, Y. He, W. Tong, J. Hua, and S. Zhong, "The fire thief is also the keeper: Balancing usability and privacy in prompts," 06 2024. [Online]. Available: https://arxiv.org/abs/2406.14318

TABLE XV

UNION RECALL DELTA (PERCENTAGE POINTS, POSITIVE RULE) WITH PAIRED GROUPED 95% INTERVAL BY SOURCE. POSITIVE ROWS ARE SHOWN AS F1/F4. N IS THE FIRST REVISION, K THE SHIPPED REWORK, O THE ORIGINAL DETECTORS. NEMOTRON-PII HAS FEW F4 ROWS AND IS REPORTED ONLY.

Source (F1/F4 rows)	F1 N−O	F4 N−O, K−O	F4 K−N
ai4privacy (3,374/2,950)	+1.96 [+0.95, +2.92]	+1.83 [+0.72, +2.97]; +2.41 [+1.32, +3.56]	+0.58 [+0.31, +0.86]
Gretel (3,325/2,863)	−0.45 [−1.30, +0.37]	−0.87 [−1.69, −0.07]; +0.52 [−0.28, +1.33]	+1.40 [+0.98, +1.85]
Nemotron-PII (1,687/682)	−4.86 [−6.36, −3.38]	−2.35 [−4.60, −0.15]; −0.73 [−2.97, +1.49]	+1.61 [+0.72, +2.69]
Privy (1,614/2,151)	−6.57 [−8.29, −4.93]	−8.04 [−9.63, −6.45]; −8.00 [−9.57, −6.41]	+0.05 [+0.00, +0.14]

[15] E. Keleş and Ş. Bahtiyar, "PromptShield: Policy-aware DLP framework for generative AI prompts," in *2025 10th International Conference on Computer Science and Engineering (UBMK)*, 2025, pp. 1537–1542, distinct conference publication record; relationship to the 2026 thesis not established. [Online]. Available: https://doi.org/10.1109/UBMK67458.2025.11206862

[16] Ú. Erlingsson, V. Pihur, and A. Korolova, "RAPPOR: Randomized aggregatable privacy-preserving ordinal response," in *Proceedings of the 2014 ACM SIGSAC Conference on Computer and Communications Security*, 2014, pp. 1054–1067. [Online]. Available: https://research.google/pubs/rappor-randomized-aggregatable-privacy-preserving-ordinal-response/

[17] H. P. Luhn, "Computer for verifying numbers," U.S. Patent 2,950,048, Aug. 23, 1960, filed 6 January 1954.

[18] International Organization for Standardization, "Financial services — International bank account number (IBAN) — Part 1: Structure of the IBAN," ISO 13616-1:2020, 2020.

[19] M. Honnibal, I. Montani, S. Van Landeghem, and A. Boyd, "spaCy: Industrial-strength natural language processing in Python," 2020, zenodo. [Online]. Available: https://doi.org/10.5281/zenodo.1212303

[20] C. Dwork and A. Roth, "The algorithmic foundations of differential privacy," *Foundations and Trends in Theoretical Computer Science*, vol. 9, no. 3–4, pp. 211–407, 2014.

[21] "PIIMB: PII Masking Benchmark," Pinned Hugging Face dataset card, revision 4a13e9f; accessed 14 October 2026. Character-level zero-shot masking benchmark; CC BY-NC 4.0.

[22] T. Heo, "Control Group v2," The Linux Kernel documentation. https://docs.kernel.org/admin-guide/cgroup-v2.html, CPU and Memory interface definitions; accessed 15 October 2026.